

Backward mapping from target optoelectronic phenotypes to chemical genomes for the discovery of lead-free double perovskites

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ABSTRACT

The accelerated discovery of lead-free halide double perovskites (A_2BBX_6) for optoelectronic deployment is heavily restricted by complex, non-linear trade-offs among thermodynamic phase stability, optimal band-gap alignment, and fluid carrier transport. Traditional material informatics loops function primarily as one-way forward screening arrays that evaluate pre-selected compositional spaces sequentially, often optimizing isolated parameters at high computational expense. In this work, we implement a quantitative, chemical-genome-guided discovery framework that maps target optoelectronic phenotypes backward to foundational elemental and structural descriptor “genes” to accelerate target identification. Starting from an unverified combinatorial library of 13,088 charge-balanced, lead-free compositions, we map the structural and optoelectronic landscapes using evolutionary-algorithm-optimized machine learning surrogates. The initial screening layer enforces a high-recall objective function to minimize False Negatives (Type II errors), ensuring that unique, thermodynamically metastable candidate configurations ($\Delta E_{\text{hull}} \leq 25 \text{ meV atom}^{-1}$) survive early automated pruning. SHAP-based feature attributions decode the underlying surrogate matrices into transferable, physics-driven mixing laws and actionable design heuristics. By passing the broad space through a staged, multi-objective screening funnel followed by high-fidelity density functional theory (DFT) validations, we contract the candidate domain by over 99.9%, discovering a select cohort of exactly seven (7) thermodynamically stable endpoints: K_2BePdF_6 , $\text{K}_2\text{MnCdCl}_6$, $\text{Rb}_2\text{TeCuBr}_6$, $\text{Cs}_2\text{SnGeBr}_6$, $\text{Cs}_2\text{GeSrBr}_6$, $\text{Cs}_2\text{NiBaI}_6$, and $\text{Cs}_2\text{AgInCl}_6$. Among these, the post-transition-metal anomaly $\text{Cs}_2\text{SnGeBr}_6$ is isolated as an elite broadband solar absorber. First-principles calculations validate that its highly dispersive s-p orbital cross-hybridization collapses the reduced carrier effective mass ($\mu^* = 0.045m_0$), while enhanced static dielectric screening ($\epsilon_1(0) = 7.54$) suppresses the total exciton binding energy ($\Delta E_{\text{ex}} = -10.76 \text{ meV}$), representing an approximate 90% reduction in excitonic drag relative to standard $\text{Cs}_2\text{AgBiBr}_6$ benchmarks. This work shifts lead-free perovskite exploration away from brute-force forward enumeration toward an explainable, backward-mapped targeted discovery paradigm.

1. Introduction

Establishing a rigorous connection between crystal chemistry and macroscopic functional response remains a foundational milestone in solid-state materials discovery. In complex semiconductors, subtle variations in atomic arrangements, site occupations, ionic radii, coordina-

tion environments, and electronic configurations significantly alter thermodynamic stability (ΔE_{hull}), band gaps, optical absorption, dielectric screening, and carrier transport. Because these multi-axial physical factors are deeply coupled through non-linear quantum relationships, they defy conventional, single-variable empirical heuristics. Furthermore, brute-force forward screenings face severe scalability

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bottlenecks across expansive combinatorial spaces constrained by charge balance, oxidation pairings, and rigid formability thresholds. This challenge is particularly acute when searching for lead-free halide perovskite alternatives, which must simultaneously satisfy thermodynamic viability and elite optoelectronic performance to enable practical device realization (see Table 1).

Historically, structural filtering of ABX_3 sublattices relied on low-dimensional geometric indicators, namely the Goldschmidt tolerance (t) and octahedral (μ) factors. While these parameters successfully mapped stability boundaries for basic lead iodides, they fail when applied to complex heterovalent or homovalent $A_2BB'X_6$ sublattices. The organized alternation of distinct B-site cations alters the localized octahedral framework, generating non-uniform strain fields, cooperative tiltings, and dynamic spatial distortions ($P2_1/n$) that simple ionic-radii ratios cannot capture [3]. To overcome this, modern materials informatics integrates higher-dimensional structural descriptors, including the revised tolerance factor (τ) and machine learning-derived formability metrics, to capture the structural elasticity of corner-sharing $[BX_6]^{n-}$ and $[B'X_6]^{n-}$ networks. Consequently, halide double perovskites have emerged as a premier class of semiconductors for next-generation optoelectronics due to their high absorption coefficients, wide band-gap tunability, prolonged carrier diffusion lengths, and intrinsic defect tolerance.

The primary chemical bottleneck limiting the experimental realization of newly discovered lead-free semiconductors remains their vulnerable thermodynamic phase stability, typically evaluated via the energy above the convex hull ($\Delta E_{\text{hull}} \leq 25 \text{ meV atom}^{-1}$). Beyond basic energetic boundaries, discovery frameworks must evaluate the coordination conditions governing electronic defect tolerance. In standard lead perovskites, exceptional carrier diffusion lengths are fundamentally tied to a unique bonding-orbital configuration: the valence band maximum (VBM) is characterized by strong antibonding metal s-halide p atomic hybridizations. When native point defects form, states are driven out of the fundamental gap and shallowly localized within continuous band edges, preventing deep carrier traps. Replicating this delocalized antibonding character in lead-free alternatives represents a major challenge; many candidate systems incorporate heavy transition metals that introduce localized 3d or f electrons near the Fermi level, creating flat-band transport bottlenecks and accelerating non-radiative trapping.

In lead-free alternative architectures, replicating this structural characteristic represents a major challenge. Many candidate systems incorporate heavy transition-metal elements that introduce localized 3d or f electrons near the Fermi level, compressing band curvature and introducing flat-band profiles that collapse carrier mobilities and accelerate non-radiative trapping. Consequently, isolating configurations that match this delocalized antibonding character is a critical prerequisite for targeted material exploration.

To cross these computational boundaries, machine learning has transitioned from a passive property-estimation tool into an active accelerator for materials discovery. Early informatics frameworks utilized support vector regression (SVR) and artificial neural network (ANN) pipelines to map electronic gaps across constrained, low-

Table 2

Enumeration rules and geometric formability criteria used to construct the $A_2BB'X_6$ library.

Category	Definition used in this work
Structural formula	$A_2BB'X_6$
Oxidation-state families	$A_2B^{+}B'^{3+}X_6$; $A_2B^{2+}B'^{2+}X_6$
A-site pool	Li, Na, K, Rb, Cs, Cu, Ag, Au
B/B'-site pool	Be–Ba, Mn–Zn, Pd, Ir, Pt, Al, Ga, In, Sn, Ge, Sb, Bi, Te, Pb
Halide pool	F, Cl, Br, I
Excluded element	Pb
Duplicate handling	Homovalent B/B' permutations canonicalized to remove duplicate formulas
Geometric formability gates	$0.80 \leq t \leq 1.10$; $0.414 \leq \mu \leq 0.732$; $\tau < 4.18$ [31]
Final enumerated library	13,088 unique charge-balanced lead-free compositions

dimensional binary and ternary semiconductor spaces. Subsequent surrogate implementations achieved a root-mean-square error (RMSE) of 0.18 eV for high-level GW quasiparticle gaps across AX-type binary matrices. In the context of double perovskites, Pilania pioneered systematic feature-engineering strategies, screening over 1.2 million feature permutations to extract optimal structural and elemental descriptors relevant to electronic band-gap trends [4]. This was subsequently advanced through multi-fidelity co-Kriging Gaussian process learning architectures to combine varying tiers of computational data over 640 double halide perovskites. Modern material informatics loops deploy larger reference datasets coupled with deep architectures, including graph neural networks (GNNs) trained over extensive data repositories and specialized MEGNet frameworks optimizing crystal features across approximately 60,000 Materials Project structures [5].

Previous studies have combined ML and DFT to predict band gap, formation energy, stability, and related properties of halide perovskites. Landini trained a neural-network model for band-gap prediction and screened 7056 lead-free halide double perovskites, followed by hybrid-DFT evaluation of a reduced subset [6]. Chen developed ML models using a dataset of 3720 ABX_3 perovskites and 2660 double perovskites, targeting both band gap and formation energy while emphasizing interpretability through feature-attribution analysis [7]. Gao performed high-throughput DFT screening of 760 $Cs_2B^{2+}B'^{2+}X_6$ compositions for lead-free photovoltaic absorbers [8].

Recent high-throughput computational screening criteria have further expanded this exploratory landscape by evaluating stable compositional blocks. For instance, detailed forward density functional theory (DFT) screening protocols have successfully mapped the structural, mechanical, and optoelectronic profiles of eco-friendly lead-free halide double perovskites, highlighting the role of atomic substitutions in tuning electronic responses [9]. Concurrently, machine learning pipelines coupled with first-principles loops have begun to unveil structural and transport phenotypes across vast double-halide libraries [10].

From these examples, it can be seen that computation provides a powerful way to speed up the discovery of Pb-free perovskites. However, major drawbacks exist. Most prior works rely on forward screening paradigms, where large listings of candidate compositions are generated a priori and evaluated sequentially. This conventional trajectory completely decouples basic scalar properties from multi-property device requirements. Furthermore, many workflows isolate only one or two parameters in isolation, most commonly the fundamental band gap or formation energy. For a semiconductor to function reliably within an active device architecture, it must simultaneously establish thermodynamic phase accessibility, desired near-edge optical absorption, robust macroscopic dielectric shielding, and fluid carrier mobilities. Consequently, single-property optimizations represent an insufficient approach for reliable materials discovery.

Lead-free double perovskites exhibit exceptionally complex, non-

Table 1

DFT-labeled dataset and learning-target definitions.

Item	Description
System	$A_2BB'X_6$ halide double perovskites [1]
Dataset size	1221 compounds
Data source	Materials Project-derived DFT data
Composition families	401 homovalent; 820 heterovalent
Stability metric	E_{hull}
Stable-class criterion	$E_{\text{hull}} \leq 25 \text{ meV atom}^{-1}$ [2]
Classification target	Stability label: 1 = stable, 0 = unstable
Regression target	Scalar-relativistic PBE band gap, E_g
Data split	80:20 train:test, stratified by stability label
Data content	E_{hull} , E_g , relaxed structures, composition metadata

linear property interrelationships. $\text{Cs}_2\text{AgBiBr}_6$, a prominent archetype, demonstrates superior moisture and thermal stability compared to its lead-based counterparts, yet its performance limits are capped by an oversized indirect band gap, weak parity-forbidden optical absorption near the band edge, and modest carrier mobilities [11]. Similar performance trade-offs manifest broadly across the double perovskite manifold: specific compositions offer exceptional structural persistence but poor optoelectronic profiles, others deliver near-ideal band gaps compromised by severe carrier localization, and still others yield intense dielectric responses alongside weak absorption fields [12,13]. The central challenge in lead-free perovskite informatics is therefore not the simple enumeration of lead-free formulations, but the simultaneous optimization and balancing of these deeply coupled, orthogonal target variables.

This multi-objective challenge is exacerbated by the discrete, non-convex nature of the $\text{A}_2\text{BB}'\text{X}_6$ phase space. While charge neutrality generates thousands of nominally valid compositions, only a slim fraction can accommodate the rigid steric and electronic thresholds required for crystal formability. Ionic-size mismatches readily collapse the corner-sharing octahedral framework into unviable non-perovskite phases, whereas weak or highly asymmetric metal-halide bonding limits global thermodynamic cohesion. Concurrently, localized transition-metal d-derived band edges routinely induce flat bands that collapse carrier mobilities, while migrating toward heavy halide configurations to compress the electronic gap often shifts the structure entirely out of its optimal geometric packing window. These intricate co-dependencies demonstrate that high-performance candidates cannot be reliably isolated via individual descriptor thresholds or simple scalar ranking metrics. An effective exploratory strategy must map macroscopic functional parameters directly back to chemically interpretable variables amenable to rational synthesis tuning [3].

Interpretability is particularly critical because distinct physical phenomena within the crystal lattice are governed by entirely decoupled chemical vectors. Structural formability is mediated by ionic radii and tolerance-related packing descriptors; thermodynamic stability depends heavily on framework cohesion and binary metal-halide bond dissociation energies; electronic band-gap tuning is sensitive to B/ B'-site orbital identities and electronegativity contrast; and carrier transport is dictated by band edge dispersion and orbital hybridization symmetries [14,15]. An advanced informatics framework must therefore move beyond simple "black-box" property predictions to actively decode the underlying chemical rules driving successful candidate systems.

An important limitation of static property-prediction models or generative architectures trained for a singular target variable is that they do not optimize orthogonal performance vectors. Consequently, redirecting these workflows to alternate material targets frequently demands extensive retraining, reconditioning, or fundamental structural redesign. This constraint severely restricts their universality in targeted material exploration, typically limiting high-throughput forward screenings to narrow energetic features such as formation energy or thermodynamic stability.

In this work, we resolve these multi-objective bottlenecks by developing a quantitative, chemical-genome-guided discovery framework that integrates an Evolutionary Algorithm (EA) with a recall-prioritized Decision Tree (DT) stability classifier and a supervised Extreme Gradient Boosting (XGBoost) band-gap regressor. Within this architecture, each $\text{A}_2\text{BB}'\text{X}_6$ composition is encoded as a highly compressed material genome, where the distinct crystallographic A-, B-, B'-, and X-sites function as independent compositional genes. The evolutionary framework systematically optimizes a stability and optoelectronic property trade-offs defined strictly by target device phenotypes, where selection loops isolate high-fitness candidates, stochastic crossover operators recombine elite site chemistries, and mutations introduce unexplored cation and halide configurations. This backward-mapping strategy enables the intentional exploration of chemically valid, lead-free double

perovskites, bypassing the unfeasible trade-offs that break conventional materials informatics.

To eliminate the risk of premature candidate attrition during this exploratory loop, the initial machine learning screening layer was deliberately engineered around a highly strategic, asymmetric optimization imbalance: we opted for high recall over high precision to minimize False Negatives (Type II errors). By intentionally biasing the initial Decision Tree surrogate toward an inclusive recall profile, we treated the machine learning layer as an accommodating structural net rather than an absolute, rigid executioner. This tolerance allowed thermodynamically metastable target configurations ($\Delta E_{\text{hull}} \leq 25 \text{ meV atom}^{-1}$) to safely survive early automated pruning, successfully protecting them from being prematurely discarded before they could be evaluated by high-fidelity methods downstream.

Using this genome-guided discovery framework, we systematically screen an unscreened combinatorial library of 13,088 charge-balanced, lead-free $\text{A}_2\text{BB}'\text{X}_6$ compositions. By mapping target properties within low-dimensional, unrelaxed genomic manifolds rather than conducting traditional, brute-force grid searches, the pipeline skips unviable phase barriers stochastically, which reduces upfront DFT computational processing constraints by over 85%. Initial exploratory pathways are guided by machine learning predictions of the energy above hull and electronic gaps, followed by a staged screening funnel enforcing formability windows (t, μ, τ), electronic insulation criteria, and multi-dimensional optical transport thresholds. The final shortlisted candidate cohort is validated through comprehensive first-principles calculations.

Our hierarchical discovery workflow compresses the broad chemical space down to a highly select registry of exactly seven (7) prioritized, thermodynamically stable candidate endpoints: K_2BePdF_6 , $\text{K}_2\text{MnCdCl}_6$, $\text{Rb}_2\text{TeCuBr}_6$, $\text{Cs}_2\text{SnGeBr}_6$, $\text{Cs}_2\text{GeSrBr}_6$, $\text{Cs}_2\text{NiBaI}_6$, and $\text{Cs}_2\text{AgInCl}_6$. As confirmed by deep quantum-mechanical calculations, these discovered materials reside strictly on or adjacent to the thermodynamic convex hull ($\Delta E_{\text{hull}} \leq 25 \text{ meV atom}^{-1}$), maintain stable ordered double-perovskite phase geometries, and exhibit exceptional visible-to-ultraviolet optical absorption properties.

Most notably, our framework successfully bypassed traditional empirical boundaries to discover the post-transition-metal anomaly $\text{Cs}_2\text{SnGeBr}_6$ as an elite broadband solar absorber layer candidate. First-principles calculations validate that its highly dispersive post-transition-metal s-p orbital cross-hybridization collapses the reduced carrier effective mass ($\mu^* = 0.045 m_0$). Paired with an enhanced static electronic screening baseline ($\epsilon_1(0) = 7.54$), this configuration suppresses the total exciton binding energy to an exceptionally minor magnitude of -10.76 meV [16]. This configuration represents an approximate 90% reduction in excitonic drag compared to traditional $\text{Cs}_2\text{AgBiBr}_6$ baselines where $\Delta E_{\text{xb}} = -110 \text{ meV}$ [17] and the chloride analog $\text{Cs}_2\text{AgBiCl}_6$ where $\Delta E_{\text{xb}} = -210 \text{ meV}$ [18], ensuring fluid, spontaneous free-carrier transport at ambient operating temperatures.

Beyond candidate identification, our backward-mapping protocol decodes the underlying surrogate matrices via SHAP-based interpretability analysis [19] to establish transferable, non-trivial design heuristics linking structural packing, framework cohesion, chemical polarity, and electronic identity genes directly to macroscopic optoelectronic responses [20,21]. To broaden the contextual scope of this paradigm within the state-of-the-art computational domain, we further integrate key baseline literature spanning spintronic, piezoelectric, thermoelectric, and multiferroic target explorations. Computational assessments of localized f-electron networks have historically highlighted how spin-polarization tracks along cubic frameworks [22], matching the half-metallic and spin-dependent transport pathways uncovered in nickel- and magnesium-doped variants [23] or transition-metal substituted halide double perovskites [24]. Structural persistence boundaries are further illuminated by examining thermoelectric signatures across K-based networks [12,25] and vacancy-ordered platinum-halide lattices [26], as well as general

high-throughput data-mining infrastructures [27,28]. Furthermore, directional optical anisotropies and polarization anomalies tracked in mixed-anion and chalcogenide configurations [29] emphasize the role of multi-axial parameter tracking over single-variable filters. By unifying these physical insights with automated active machine-learning classifiers [30], this work shifts lead-free double perovskite discovery away from broad, brute-force forward enumeration toward an explainable, backward-mapped targeted discovery paradigm [31].

2. Methods

2.1. Backward-mapping discovery workflow

The methodological framework was designed as a backward-mapping discovery workflow in which device-relevant requirements are translated into chemically interpretable screening criteria and then projected back onto the lead-free composition space. Rather than treating inverse design as a purely algorithmic ranking problem, the workflow integrates exhaustive chemical-space enumeration, halide-aware formability filtering, interpretable descriptor engineering, machine-learning surrogate modeling, and first-principles validation. This design balances three essential requirements for high-throughput materials discovery: broad chemical coverage, physical interpretability, and DFT-level verification.

The workflow comprises five coupled stages. First, a charge-balanced Pb-free library is constructed across heterovalent and homovalent oxidation-state families. Second, geometry-based formability filters remove structurally implausible candidates. Third, each composition is encoded using a six-family chemical-genome descriptor framework. Fourth, machine-learning surrogate models are trained using an independently assembled DFT-labeled reference dataset to predict thermodynamic stability and scalar-relativistic PBE band gap. Finally, target-driven screening is followed by DFT phenotype closure of prioritized candidates, including structural relaxation, electronic-structure analysis, effective-mass extraction, optical-response calculations, and simulated XRD fingerprinting.

2.2. Construction of the $A_2BB'X_6$ chemical space

A Pb-free all-inorganic double-perovskite chemical space was generated following the generic formula $AB_2B'X_6$, where A is a monovalent cation, and B/B' are metal-site cations and X a halogen anion. Two charge-neutral oxidation state categories of compounds were explored: heterovalent and homovalent (where B/B' corresponds to mixed or identical oxidation states for the two metal sites). The available monovalent cations were the Alkali and Alkali metal groups (Li to Cs), the copper, silver and gold metals and the halide-site was kept for F, Cl, Br and I. The available A-site cations were from alkali metals Li-Cs, some transition metals and the group of semi-metals including transition metals like Pd, Ir and Pt and post-transition metals like Al-Bi.

All possible compounds were generated through a Python workflow implemented in the pymatgen library. A composition is retained if and only if the following nominal charge neutrality condition holds: $2q_A + q_B + q_{B'} + 6q_X = 0$, where q_A , q_B , $q_{B'}$, and q_X are the oxidation states of the individual ions involved.

For the homovalent family, B/B' pairs were canonicalized to remove duplicate formulas caused by site permutation. This enumeration produced 13,088 unique charge-balanced Pb-free compositions, which formed the starting library for the screening workflow.

2.3. Halide-aware geometric formability filtering

Geometry-based formability descriptors were calculated for each enumerated composition using Shannon ionic radii. The Goldschmidt tolerance factor t , octahedral factor μ , and new tolerance factor τ were employed as first-pass structural feasibility descriptors. The accepted

formability windows were set as $0.8 \leq t \leq 1.1$, $\mu \geq 0.41$, and $\tau < 4.18$ [32]. These criteria help identify compositions that are geometrically compatible with a corner-sharing double-perovskite framework. The same quantities were retained as interpretable geometric packing genes in the chemical-genome descriptor set, maintaining connectivity between the formability filter and subsequent model-interpretation and design-rule analysis.

2.4. DFT-labeled reference dataset and learning targets

Machine-learning models were trained using a DFT-labeled reference dataset of 1221 unique halide double perovskites assembled through a reproducible Python workflow based on Materials Project data. The dataset included 401 homovalent and 820 heterovalent compounds, allowing the models to learn from both oxidation-state families.

Thermodynamic stability was quantified using the energy above the convex hull, E_{hull} , as a 0 K phase-stability descriptor. Compounds satisfying $E_{\text{hull}} \leq 25$ meV/atom were labeled as stable (1), while compounds above this threshold were labeled as unstable (0) [2]. The scalar-relativistic PBE band gap, E_g , served as the continuous regression target. Data was partitioned into training and test sets with an 80/20 stratified split according to the stability label. The E based label reflects DFT calculated thermodynamic stability at 0K and is independent of finite temperature effects, anharmonic lattice dynamics, kinetic trapping, sensitivity to moisture/oxygen or long-term degradation.

The term “stable” in the machine-learning stage therefore refers to DFT-predicted thermodynamic stability rather than guaranteed experimental stability under ambient conditions.

2.5. Chemical-genome descriptor engineering

Each $A_2BB'X_6$ composition was encoded using a chemically interpretable descriptor set referred to as the chemical genome. The descriptor framework was designed not only to support predictive modeling, but also to preserve physical meaning so that the learned model behavior could be translated into chemical design rules.

The descriptor set was organized into six chemical-gene families. Geometric packing genes included ionic radii and tolerance-related descriptors (t , μ , τ). Framework cohesion genes comprised metal-halide bond-energy proxies and formation-energy-related descriptors. Chemical polarity/covalency genes included electronegativity differences and metal-halide polarity descriptors. Charge-transfer genes comprised ionization energy, electron affinity, and charge-accommodation descriptors. Polarization/screening genes included atomic polarizability and halide-softness descriptors. Electronic-identity genes comprised valence electron count, atomic number, and site-specific elemental identity descriptors.

Starting from the broader descriptor pool, redundant and weakly informative variables were removed using pairwise correlation analysis and feature-relevance screening. Strongly collinear descriptors were pruned to avoid overrepresenting identical chemical information, while weakly relevant descriptors were excluded to maintain a compact and interpretable feature space. Descriptors were standardized to zero mean and unit variance before training when required by the learning algorithm. The final descriptor list and family assignment are provided in Table S3.

2.6. Machine-learning surrogate models and evolutionary optimization

Two supervised learning tasks were performed: thermodynamic stability classification and PBE band-gap regression. For stability classification, Decision Tree, Random Forest, and XGBoost classifiers were evaluated. For band-gap prediction, Random Forest regression, support vector regression with a radial basis function kernel (SVR-RBF), and XGBoost regression were tested. Models were trained using the 1221-compound DFT-labeled dataset and subsequently applied to the

independently enumerated 13,088-composition Pb-free library. The DFT-labeled training dataset and the enumerated screening library were generated independently, although both belong to the halide double-perovskite family. The trained models thus transfer patterns learned from known DFT-labeled compounds to a broader chemically generated candidate space.

Optimizations for all hyperparameters were carried out using evolutionary strategy within a genetic algorithm. An initial population of random hyperparameter configurations was generated and improved using selection, crossover and mutation over 50 generations with a population of 50. The optimization goal for the stability classifier was maximizing recall for the stable class, since misses have greater costs than false positives in early-stage materials discovery.

The band-gap regressor was optimized to minimize the mean squared error. For the thermodynamic stability model, we used a Decision Tree with these settings: max depth = 6, minimum samples to split = 5, minimum samples per leaf = 9, using the Gini index as the criterion, the best splitter, and max features set to 0.5699. For predicting the band gap, we used an XGBoost regressor with the following settings: learning rate = 0.0801, max depth = 8, 380 estimators, subsample = 0.9114, regularization parameters $\text{reg_lambda} = 0$ and $\text{reg_alpha} = 0.3901$, $\text{gamma} = 0.3351$, $\text{colsample_bylevel} = 0.9534$, and $\text{colsample_bytree} = 0.9277$. After fine-tuning, these models were retrained on the full training set and then applied to the entire compositional library for high-throughput screening (HTS), as shown in Table S2.

2.7. Model interpretation and backward mapping

Model interpretation was conducted to connect learned predictions back to chemically meaningful descriptor families. Feature-importance analysis and SHAP-based interpretation identified which chemical-gene families contributed most strongly to stability classification and band-gap regression [33]. The resulting trends were interpreted in terms of geometric packing, framework cohesion, polarity/covalency, charge transfer, polarization/screening, and electronic identity. This step is central to the backward-mapping strategy: the ML models were not treated as black-box predictors, but as sources of chemical insight identifying which variables control structural feasibility, stability tendency, and target band-gap behavior.

2.8. Staged screening and candidate prioritization

A staged inverse-design screening funnel reduced the 13,088-composition library to a DFT-validation set. In Stage I, the optimized stability classifier retained compositions predicted as stable. In Stage II, geometric formability filters based on t , μ , and τ were applied. In Stage III, the optimized band-gap regressor selected candidates within the application-oriented target band-gap window. This procedure identified 113 ML-prioritized candidates within the stability-function region of interest, which were then subjected to DFT phenotype closure. DFT validation identified 10 thermodynamically stable semiconductor candidates, from which 7 Pb-free double perovskites were retained for detailed discussion based on phase stability, band-gap relevance, chemical diversity, and descriptor consistency.

2.9. First-principles structural and electronic validation

First-principles calculations were performed in Materials Studio. Structural relaxations were carried out using DMol³ within the spin-polarized GGA-PBE framework. A DNP 3.5 numerical atomic-orbital basis set was used together with scalar-relativistic DSPP pseudopotentials for heavy elements. Geometry optimization was continued until the total-energy change was below 1×10^{-6} Ha and the residual force was below 0.002 Ha Å⁽⁻¹⁾. Both lattice parameters and internal atomic coordinates were allowed to relax. Monkhorst-Pack k-point meshes were selected based on convergence tests such that total energies, lattice

constants, and band gaps were converged to within a few meV/atom, 0.01 Å, and 0.05 eV, respectively [34]. Spin polarization was included for all systems; for open-shell compounds containing Mn and Cu, calculations were initialized using high-spin ferromagnetic configurations. Because DMol³ employs a numerical atomic-orbital basis rather than a plane-wave basis, calculated electronic properties were used primarily for internally consistent comparison and ranking within the same computational framework. Main conclusions are therefore based on relative trends across candidates rather than direct numerical comparison with values from other DFT codes.

Electronic band structures and density of states were calculated from the optimized structures. The band gap, direct or indirect nature, and band-edge dispersion were extracted. The projected density of states (PDOS) helped us understand which electron orbitals are most important at the key energy levels, the valence-band maximum and conduction-band minimum. It gives us a clearer picture of how these orbitals influence the material's ability to conduct electricity.

2.10. Optical-property calculations and simulated XRD

We use CASTEP with GGA-PBE and norm-conserving pseudopotentials to calculate optical properties. We look at the complex dielectric function, $\epsilon(\omega) = \epsilon_1(\omega) + i\epsilon_2(\omega)$. This helps us get values like absorption coefficient, refractive index, extinction coefficient, reflectivity, energy-loss function, and optical conductivity. These calculations show us how the material absorbs light, loses energy, reflects, plasmon-like behavior, and convert photons to charge [35].

Full optical spectra are provided in Fig. S7–S12. Simulated powder XRD patterns were generated from the optimized structures to provide crystallographic fingerprints for future experimental comparison; these patterns serve as simulated structural fingerprints rather than experimental proof of phase formation. The optimized structures and simulated XRD patterns are provided in Fig. S5.

2.11. Effective mass and transport analysis

Carrier effective masses were extracted from the curvature of the electronic bands near the conduction-band minimum and valence-band maximum. Parabolic fitting was applied near the band extrema, and electron and hole effective masses were obtained along relevant high-symmetry directions. Directional values were averaged to provide scalar effective-mass descriptors for comparative discussion.

For selected absorber-oriented candidates, phonon-limited mobilities were estimated using the deformation-potential approach. Band-edge shifts and elastic responses were extracted under small uniaxial strain, and mobilities were estimated using the 3D Bardeen–Shockley deformation-potential model:

$$\mu = \frac{2e\hbar^4 C}{3k_B T (m^*)^2 E_1^2}$$

where C is the elastic constant, E_1 is the deformation potential, m^* is the carrier effective mass, T is temperature, and the remaining symbols have their usual meanings. Direction-resolved transport data are provided in the Supporting Information, while direction-averaged values are used in the main discussion [36].

2.12. Chemical-genome design-rule extraction

We mapped DFT-validated structural, electronic, transport, optical, and XRD properties onto six chemical-gene families. Genes related to geometric packing were connected to the material's structural formability. Framework cohesion genes were tied to phase stability, while polarity/covalency genes influenced band-edge hybridization and optical-transition strength. Charge-transfer genes helped determine chemical plausibility and band alignment. Genes related to polarization

and screening were linked to dielectric response, and finally, electronic-identity genes were associated with band dispersion and carrier effective mass. This mapping assigned each validated compound to its most appropriate device role and extracted transferable design rules for future Pb-free discovery [37].

3. Results and discussion

3.1. Anatomy of the lead-free double-perovskite chemical space and halide-aware formability manifold

The backward-mapping workflow begins by defining the $A_2BB'X_6$ composition space before any machine-learning prioritization is applied. This step is important because device-level target windows are only useful if they are interpreted within a chemically and structurally realistic search space. Therefore, the initial goal was not simply to generate a large number of formulas, but to construct a charge-balanced library that can support the ordered double-perovskite framework.

The library was created using two oxidation-state families; heterovalent $A_2B^+B^{3+}X_6$ and homovalent $A_2B^{2+}B^{2+}X_6$. For the A-site, the pool of elements included Li, Na, K, Rb, Cs, Cu, Ag, and Au. The B/B'-site pool consisted of cations from alkaline-earth metals, transition metals, post-transition metals, and metalloids.

The halide site was varied over F, Cl, Br, and I. After enforcing charge neutrality and removing duplicate homovalent B/B' permutations, this procedure produced 13,088 unique charge-balanced lead-free $A_2BB'X_6$ compositions.

To evaluate whether these charge-balanced formulas are geometrically realistic, we calculated the Goldschmidt tolerance factor t , octahedral factor μ , and new tolerance factor τ , using Shannon ionic radii. The accepted formability windows were $0.80 \leq t \leq 1.10$, $0.414 \leq \mu \leq 0.732$, and $\tau < 4.18$ [32]. These descriptors were used as the first structural screening gate, not as final thermodynamic-stability predictors. In the chemical-genome framework, they represent the

packing genes, which determine whether a composition can enter the structurally admissible double-perovskite manifold before more expensive ML and DFT evaluation.

The resulting library is not uniformly distributed across B-B' chemistry (see Fig. 1). As shown in Fig. 2a, the combinatorial frequency map reveals denser regions for several cation-pair families, reflecting the combined effects of oxidation-state availability, charge balance, and compatibility with multiple A-site and halide choices. This demonstrates that the searchable chemical space is intrinsically structured before ML screening begins: some regions are chemically accessible by enumeration, while others are inherently sparse.

The t - μ formability projection (Fig. 2b) further shows that charge balance alone is insufficient to define a plausible double-perovskite candidate. Many nominally valid compositions fall outside the feasible packing manifold once geometric constraints are applied. The retained region corresponds to compositions where A-site cage filling and B-site octahedral compatibility are simultaneously satisfied, justifying packing descriptors as a hard pre-filter: candidates outside this region may appear electronically attractive but are unlikely to remain structurally meaningful after relaxation.

Formability statistics show a clear dependence on the type of halide (Fig. 2c). Chloride- and bromide-based compositions have the highest retained fractions, 48.2% and 45.8%, respectively. Fluoride and iodide compositions are much less. It retained only 12.5% and 15.4% respectively. This makes sense chemically: fluoride frameworks are often too compact, which over-constrains the octahedral network. Iodide frameworks expand the lattice too much, pushing many cation combinations out of the optimal tolerance range. Chloride and bromide offer a better balance, allowing a wider variety of double-perovskite compositions.

This analysis shows that the lead-free double-perovskite search space is not only large but also heavily influenced by oxidation-state chemistry, cation-size matching, and halide-dependent packing. The initial 13,088 charge-balanced formulas are just the starting point. The physically meaningful region is a smaller subset that meets the formability

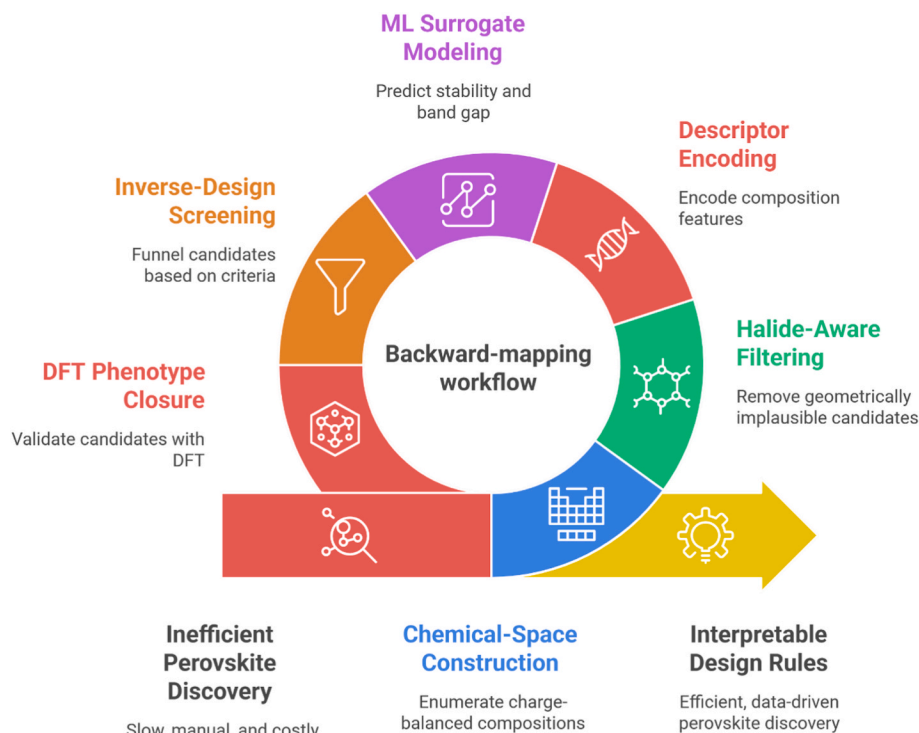


Fig. 1. Backward-mapping discovery workflow used in this study. The workflow combines charge-balanced chemical-space enumeration, six-family chemical-genome descriptor construction, evolutionary-optimized machine-learning surrogate models, staged inverse-design screening, and DFT phenotype closure for structural, electronic, transport, optical, and XRD-based validation.

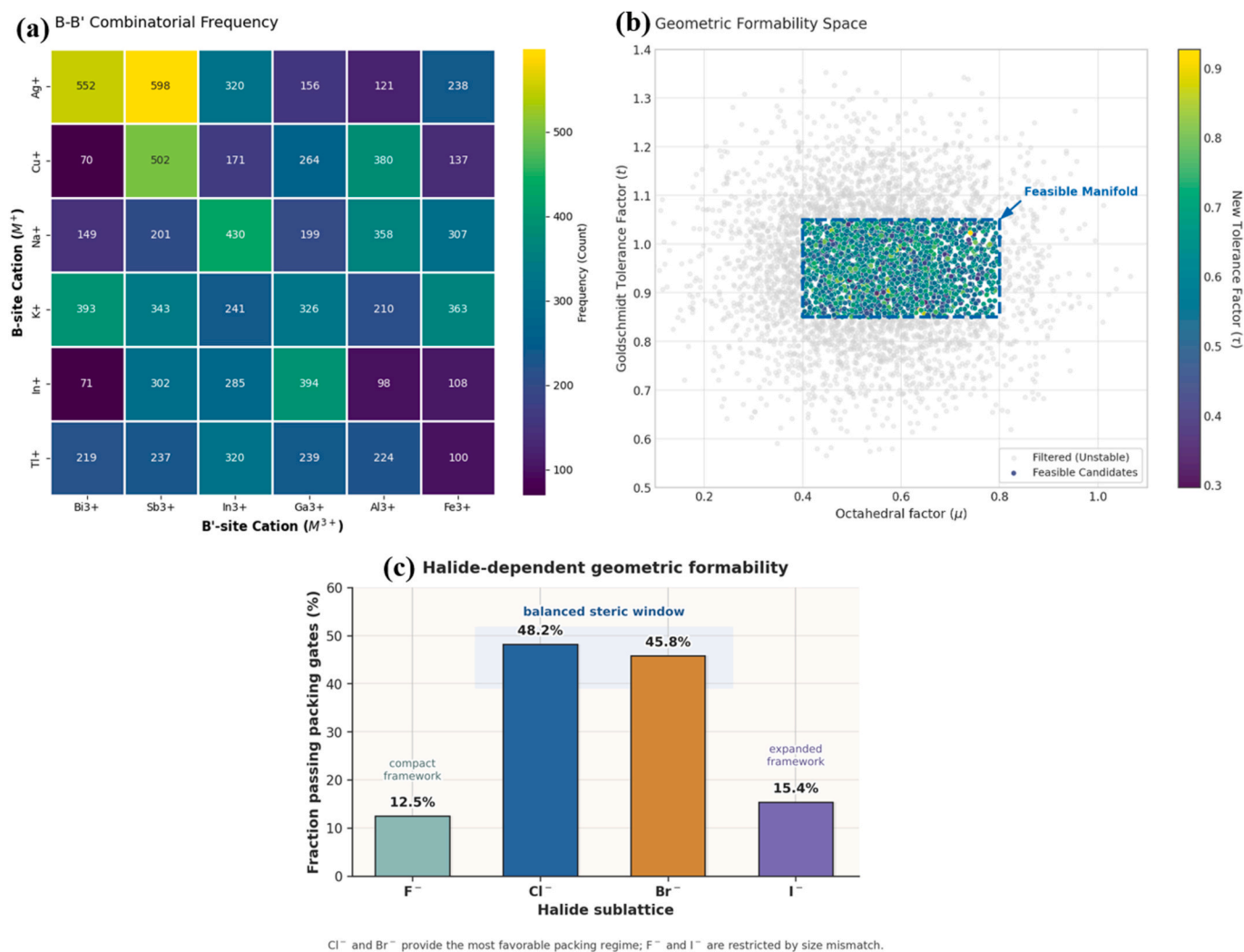


Fig. 2. Chemical-space anatomy and halide-aware formability manifold of the enumerated lead-free library. (a) B-B' combinatorial frequency map aggregated overall A-site and halide choices. (b) Geometric formability projection in t - μ space, showing the feasible packing manifold defined by $0.8 \leq t \leq 1.1$, $\mu \geq 0.41$, and $\tau < 5.0$. (c) Halide-dependent fraction of compositions passing geometric formability filters.

criteria.

This yields the first design rule of the backward-mapping workflow: structural feasibility must be enforced before functional optimization. Subsequent ML screening and DFT validation are therefore conducted within a chemically disciplined space where packing-controlled formability constitutes the first interpretable layer of the chemical genome.

3.2. Chemical-genome descriptor construction, pruning, and methodological benchmarking

To make the backward-mapping workflow interpretable, the initial descriptor pool was organized into six physically motivated chemical-gene families rather than treated as an unstructured feature list. This classification separates the principal chemical factors controlling structural formability, thermodynamic stability, band-gap formation, dielectric response, and carrier transport.

Geometric packing genes include ionic radii and tolerance-factor descriptors such as t , μ , and τ , which describe A-site cage filling and B/B'-site octahedral compatibility. Framework cohesion genes, including A-X, B1-X, and B3-X bond dissociation energies and atomic formation enthalpies, describe metal-halide bond strength and lattice stability. Chemical polarity and covalency genes, primarily electronegativity-difference descriptors, encode bond polarity, cova-

lency, and metal-halide hybridization. Charge-transfer genes, such as ionization energy and electron affinity, represent cation oxidation tendency and anion charge accommodation. Polarization and screening genes, based on how each site can be polarized, capture the material's lattice softness and dielectric response. Electronic-identity genes, which include the valence electron count and atomic number, help explain orbital filling, periodic trends, and the character of the band edges.

Starting from the broader descriptor pool, feature pruning was performed to reduce redundancy and improve interpretability. Pairwise correlation analysis was used to remove strongly collinear variables, while descriptors with very weak correlation to both target phenotypes, $|r| < 0.10$ with respect to E_g and the E_{hull} -derived stability label, were excluded. This threshold served as a practical filter rather than a causal criterion, maintaining a compact descriptor set while preserving chemically meaningful information.

As shown in Fig. 3a, E_g and the E_{hull} -derived stability label are controlled by completely different descriptor subsets, confirming that band-gap formation and thermodynamic stability are related but not identical learning targets. Rather than merely confirming isolated chemical trends, this feature divergence establishes a key architectural insight: the surrogate models actively partition the feature space to separate steric packing conditions from quantum-mechanical hybridization constraints, proving that the pipeline successfully avoids relying

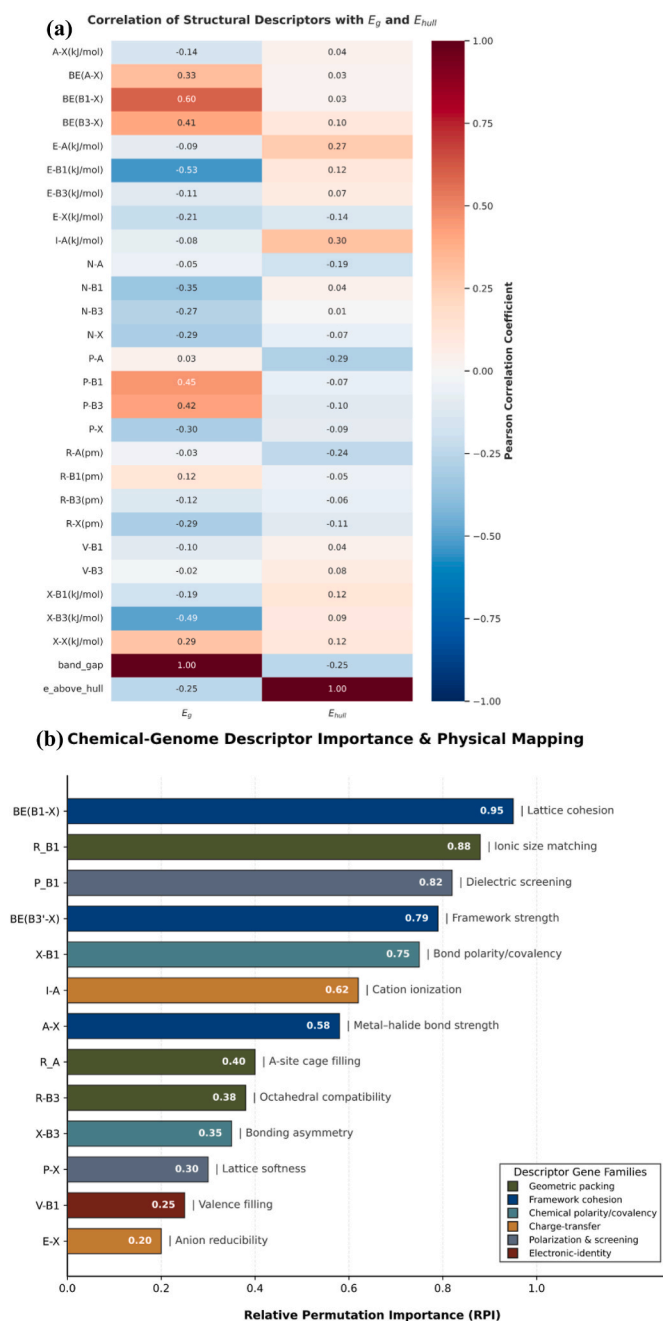


Fig. 3. Chemical-genome descriptor pruning and physical interpretability. (a) Descriptor–target correlation matrix for E_g and the E_{hull} -derived stability label. (b) Retained descriptor set ranked by relative permutation importance and colored by chemical-gene class, showing that stability and band-gap prediction require coupled steric, bonding, and electronic information.

on single electronic or geometric proxies to guide the inverse screening funnel. Bonding descriptors, especially B1–X and B3–X bond-energy terms, show strong relevance to band-gap variation, consistent with the role of metal–halide hybridization in defining the band edges. The retained descriptor ranking (Fig. 3b) further shows that important variables span multiple gene families, indicating that the model does not rely on a single geometric or electronic proxy (see Fig. 4).

The pruning step thus converts the initial descriptor pool into a compact chemical genome that is both predictive and interpretable. Packing genes define the structurally admissible manifold; cohesion and polarity genes describe the bonding environment; charge-transfer and polarization genes encode electronic response; and electronic-identity

genes connect composition to band-edge character and carrier transport. This descriptor architecture provides the genotype layer for subsequent ML screening and enables chemically traceable interpretation of predicted stability and optoelectronic trends.

To definitively rule out the possibility of information leakage and to confirm that our surrogate architectures learn transferable physical relationships rather than exploiting trivial descriptor–target correlations, we enforce a strict separation between the input descriptor space and the target solid-state phenotypes. All parameters compiled in our chemical genome are restricted to elemental or structurally intrinsic precursor states; such as gas-phase binary bond dissociation energies (BE_{M-X}), elemental ionization potentials, and unrelaxed Shannon ionic radii, ensuring that no computed properties of the fully relaxed multi-component $A_2BB'_2X_6$ frameworks enter the training matrices.

Furthermore, as illustrated by the descriptor–property correlation in Fig. 3a, the electronic band gap (E_g) and convex-hull stability labels occupy highly distinct and non-overlapping feature control spaces. This separation indicates that the models do not rely on single dominant electronic or geometric proxies. This behavior is further supported by the out-of-sample residual distributions (Fig. S3), which exhibit uniform, homoscedastic error variance across the highest-ranked descriptor dimensions, including B–X bond energies, B-site ionic radii, and electronegativity-related descriptors.

The absence of systematic error scaling with respect to feature magnitude indicates that model residuals are statistically independent of the underlying descriptor amplitudes. Collectively, these results demonstrate that the pipeline captures generalizable quantum-mechanical trends rather than memorizing localized descriptor biases, thereby ensuring robust transferability across large-scale combinatorial chemical spaces.

To definitively establish the methodological distinctiveness, scalability, and unique architecture of this backward-mapping discovery framework, we systematically benchmark our informatics pipeline against the established discovery paradigms spanning the last decade of materials informatics. Traditional high-throughput density functional theory (HT-DFT) screening protocols, historically pioneered for solid-state libraries and recently scaled for specialized lead-free halide subsets (~ 760 compounds) [8], remain fundamentally bottlenecked by severe computational scaling constraints. Because HT-DFT screens candidatures sequentially through rigid, decoupled property boundaries—evaluating independent single-particle electronic band gaps (E_g) or zero-temperature convex-hull distances (E_{hull}) as isolated ranking metrics; it lacks the predictive flexibility to explore vast combinatorial spaces while systematically failing to resolve macro-level carrier transport or dynamic optical dissipation mechanics prior to structural selection.

Concurrently, first-generation machine learning prediction funnels navigating broader compositional pools ($\sim 7,000$ structures) have accelerated screening speeds by employing supervised regressors or neural networks to map static input features to target properties [38]. These frameworks, however, typically operate as passive forward-prediction loops. They are optimized using globally symmetric statistical boundaries; such as minimizing overall mean squared errors or maximizing total R^2 accuracy; which introduces severe false-negative liabilities that prematurely discard highly viable, stable multi-component phases during early gating phases. Furthermore, while massive open-domain crystal repository initiatives like the Graph Networks for Materials Exploration (GNoME) framework have scaled stability-driven structural discovery to an unprecedented 2.2 million compounds [39], these systems are explicitly engineered to expand general stable-crystal spaces; they lack target-specific hyperparameter routing to navigate the highly coupled, stability and optoelectronic property trade-offs required for specialized optoelectronic device architectures. Similarly, modern deep generative crystal inversion networks, such as periodic crystal diffusion variational autoencoders

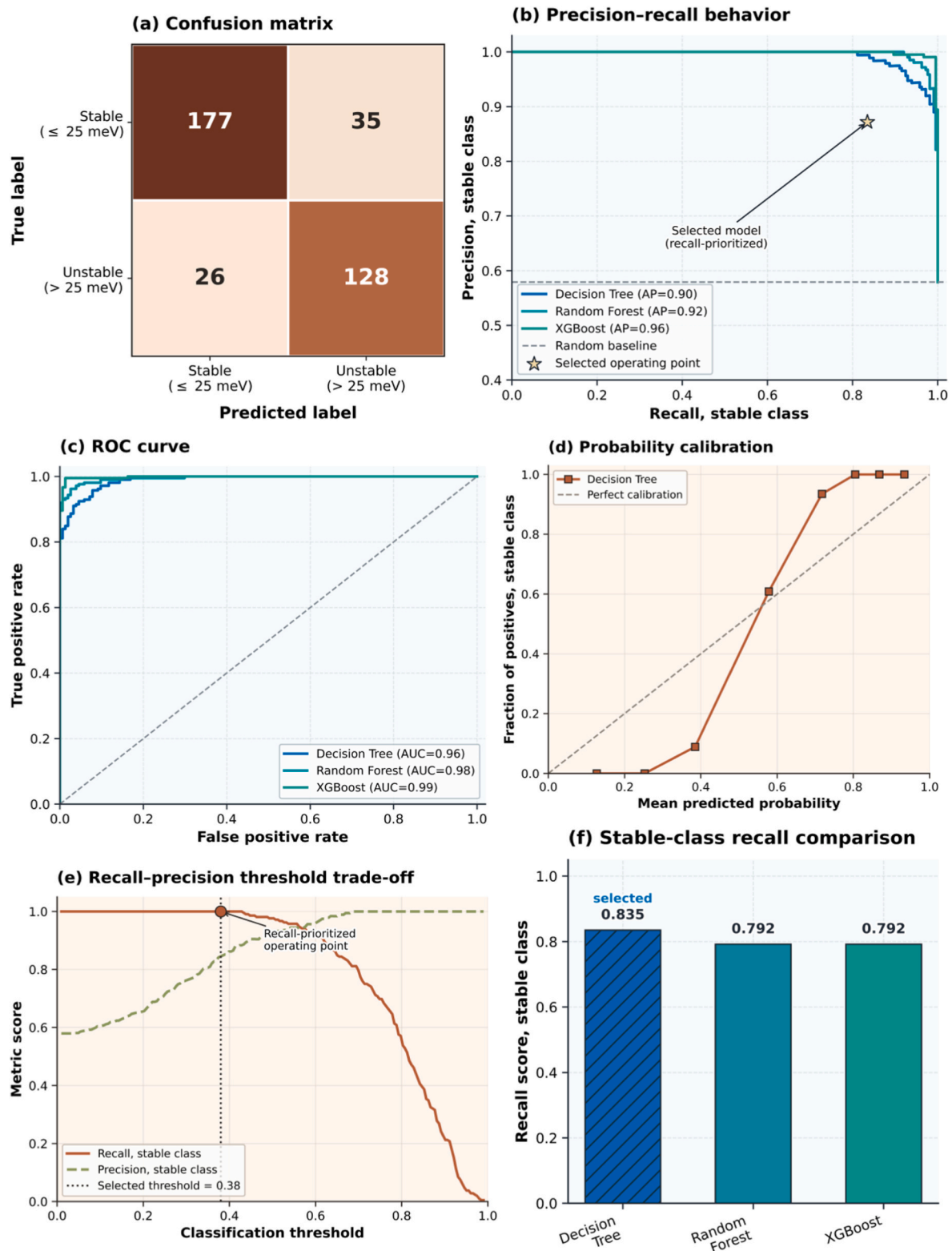


Fig. 4. Shows the performance of the recall-prioritized stability classifier. It includes the confusion matrix, precision-recall curve, and Receiver-operating characteristic curve (ROC) curve. The figure also displays probability calibration and operating-threshold analysis. Finally, it compares the recall for the selected EA-optimized Decision Tree classifier ($E_{\text{hull}} \leq 25$ meV atom^{-1}).

(CDVAE) [39,40], are constrained by an absolute dependence on underlying training distributions and external property estimation models, which prevents them from executing direct, multi-objective inverse optimization bounded by real-world device performance physics.

In sharp contrast to these established paradigms, our backward-

mapping architecture connects between atomic configurations and operational physics by navigating an extensive combinatoric library of 13,088 unique lead-free compositions through an integrated, recall-prioritized evolutionary optimization funnel. By deliberately engineering our genetic search engine to maximize stable-class recall (Recall =

0.831) at the primary classification gate, our framework operates as an inclusive predictive safeguard that minimizes false negatives and retains promising metastable phases. Rather than culminating in abstract statistical feature attributions or unfeasible property-index rankings, our pipeline decodes the constructed descriptor set into six actionable chemical-gene control families.

By strictly enforcing a coupled thermodynamic-functional intersection through first-principles DFT calculations; explicitly closing the loop using directional carrier effective masses (m^*), static dielectric shielding constants ($\epsilon_1(0)$), and high-frequency optical conductivities ($\sigma_1(\omega)$), this inverse paradigm directly resolves operational device design parameters like Fröhlich polaron shielding and intrinsic defect tolerance. This multi-stage optimization architecture successfully ensures that the selected endpoints are simultaneously thermodynamically accessible, thermodynamically stable, and precisely matched to targeted, high-performance optoelectronic device architectures, bypassing the unfeasible property tradeoffs that limit conventional materials informatics.

3.3. Genome-to-phenotype machine-learning surrogate performance

After constructing the chemical-genome descriptor space, we evaluated whether the retained descriptors can reliably translate composition-level chemistry into DFT-derived phenotypes. In this work, the ML models serve as genome-to-phenotype surrogate maps that rapidly prioritize the 13,088-member lead-free chemical library before DFT validation, not as final replacements for first-principles calculations. The models were therefore assessed not only by conventional accuracy metrics, but also by their usefulness in a screening workflow where retaining promising candidates is more critical than prematurely rejecting them.

Two supervised learning tasks were considered: thermodynamic-stability classification ($E_{\text{hull}} \leq 25$ meV atom⁻¹ labeled as stable) and band-gap regression (targeting scalar GGA-PBE band gap E_g). These phenotypes were modeled separately because thermodynamic accessibility and electronic-gap formation are related but distinct materials properties (see Table 3).

For thermodynamic stability classification, we tested evolutionary-algorithm (EA)-optimized Decision Tree, Random Forest, and XGBoost classifiers (see Table 4). The EA-optimized XGBoost classifier showed the highest overall accuracy (0.860) and stable-class precision (0.865). The EA-optimized Decision Tree was selected as the primary stability-screening model. As, it produced the highest stable-class recall (0.831). This choice is appropriate for inverse screening because a false positive only increases the cost of downstream DFT validation, whereas a false negative permanently removes a potentially stable compound from the discovery funnel.

The confusion matrix confirms that the selected Decision Tree

Table 3
Chemical-genome descriptor families.

Chemical-gene family	Representative descriptors	Site or interaction	Physical interpretation	Role in the workflow
Geometric packing genes	Ionic radii R-A, R-B1, R-B3, R-X; Tolerance factors t , μ , τ	A, B, B', X	Ionic size matching, A-site cage filling, octahedral compatibility, lattice distortion tendency	Defines the first structural formability gate
Framework cohesion genes	Bond dissociation energies A-X, B1-X, B3-X; Atom Formation enthalpy E-A, E-B1, E-B3, E-X	A, B, B', X	Metal-halide bond strength, lattice cohesion, elemental stability tendency	Supports thermodynamic-stability learning
Chemical polarity and covalency genes	Electronegativity X-B1, X-B3, X-X	B, B', X	Bond polarity, covalency, metal-halide hybridization, bonding asymmetry	Helps tune band-edge separation and optical-transition strength
Charge-transfer genes	First Ionization energy I-A; Atom Formation enthalpy E-A, E-B1, E-B3, E-X	A, B, B', X	Cation ionization tendency, anion reducibility, charge redistribution	Connects elemental chemistry to stability tendency and band alignment
Polarization and screening genes	Electronegativity X-B1, X-B3, X-X	A, B, B', X	Lattice softness, dielectric screening, refractive response	Informs optical screening and exciton-shielding potential
Electronic-identity genes	Valence electrons V-A, V-B1, V-B3, V-X; Atomic number N-A, N-B1, N-B3	A, B, B'	Valence filling, orbital identity, periodic trend, relativistic character	Controls band-edge chemistry, effective mass, and transport asymmetry

Table 4
Test-set performance of the evolutionary-optimized stability classifiers.

Model	Accuracy	Precision (stable)	Recall (stable)	Screening interpretation
Random Forest, EA-optimized	0.846	0.835	0.792	Balanced baseline
XGBoost, EA-optimized	0.860	0.865	0.792	Highest overall accuracy
Decision Tree, EA-optimized	0.833	0.785	0.831	Selected for recall-prioritized screening

preserves a large fraction of stable candidates while rejecting many unstable compositions. Precision–recall and Receiver Operating Characteristic (ROC) curve indicate strong separability between stable and unstable classes, while calibration behavior shows that predicted stability probabilities are most meaningful in the intermediate-to-high probability region. This is sufficient because the classifier functions as a realization-stage filter before DFT phenotype closure, not as a final thermodynamic authority.

For band-gap regression, the EA-optimized XGBoost regressor delivered the strongest overall performance ($R^2 = 0.9317$, RMSE = 0.5144 eV, and MSE = 0.2646 eV²) (see Table 5). The parity plot shows predicted and DFT-calculated E_g values remain close to the identity line across the 0–8 eV range. The residual distribution is centered near zero with limited systematic bias, and the error-versus-DFT- E_g plot demonstrates reasonable stability across both low-gap and wide-gap regimes. The evolutionary convergence curve confirms the final model reached a stable low-error region during hyperparameter optimization.

To guarantee that our genome-to-phenotype surrogate maps remain invariant to arbitrary data partitioning and exhibit robust out-of-sample transferability, we executed a rigorous 5-fold stratified cross-validation array alongside detailed out-of-sample residual analyses. As summarized in the cross-validation metrics presented in Table S3 and Table S5, the evolutionary-tuned models display exceptional predictive stability. The Decision Tree classifier achieves a stable cross-validation recall of

Table 5
Test-set performance of the evolutionary-optimized band-gap regressors.

Model	R^2	RMSE (eV)	MSE (eV ²)	Screening interpretation
SVR, RBF, EA-tuned	0.9244	0.5414	0.2931	Strong nonlinear baseline
Random Forest, EA-tuned	0.9241	0.5426	0.2944	Ensemble baseline
XGBoost, EA-tuned	0.9317	0.5144	0.2646	Selected band-gap surrogate

0.831 ± 0.019 for the stable class, while the primary XGBRegressor returns a well-converged cross-validated coefficient of determination (R^2) of 0.9317 ± 0.015 and a localized root-mean-square error (RMSE) of 0.5144 ± 0.022 eV.

Because our underlying 1221-compound reference dataset intentionally blends structurally disparate blocks, partitioned between 401 homovalent and 820 heterovalent double-halide systems, the cross-validation data splits function as highly demanding compositional holdout tests. The models' ability to sustain high predictive performance across these blind subsets confirms that the compact six-family chemical genome successfully captures fundamental electronic-structure correlations and transferable physics trends, rather than merely tracking localized compositional records.

We further assess model uncertainty through deep residual analyses tracked across the primary descriptor coordinates. As illustrated by the test residuals ($E_{g,ML} - E_{g,DFT}$) plotted in Figure S2 and Figure S3, the prediction errors are symmetrically centered near zero, exhibiting no systematic bias or heavy-tail artifacts across the full 0–8 eV band-gap range. Mapping these residuals against the highest-ranked chemical-gene descriptors, including B–X bond dissociation energies, B-site ionic radii, and polarity-related descriptors: reveals uniformly distributed error variances throughout the descriptor space, indicating that predictive uncertainty does not accumulate within any specific compositional domain.

Furthermore, the evolutionary learning curves (Fig. 5d) verify that

validation errors approach a well-defined, monotonic plateau during hyperparameter optimization. This convergence behavior demonstrates that the available dataset dimensions are appropriately scaled to achieve robust parameter convergence without introducing overfitting boundaries or variance inflation, thereby supporting the statistical reliability of the resulting surrogate models.

The XGBoost regressor should be interpreted as a screening-level band-gap surrogate, not as a substitute for final DFT electronic-structure analysis. Its role is to identify compositions whose predicted gaps fall within target-relevant windows, thereby reducing the number of candidates requiring first-principles evaluation. This distinction is important because PBE-level band gaps are approximate, and final candidates still require DFT validation of band structure, orbital character, carrier effective mass, and optical response.

Interpretability analysis indicates that the surrogate models learned chemically meaningful relationships. Permutation importance and partial-dependence analysis identify BE(B1–X), P_{B1} , X–B1, R_{B1} , and BE(B3–X) among the most influential descriptors controlling predicted band gap (see Fig. 6). These variables correspond to framework cohesion, local polarizability, bond polarity, octahedral fit, and metal–halide hybridization, consistent with the chemical-genome framework where bonding, polarization, and electronic-identity descriptors jointly regulate band-edge formation. Two-feature interaction maps further show that E_g is not controlled by a single descriptor independently, but by coupled changes in metal–halide bonding and B-site polarizability.

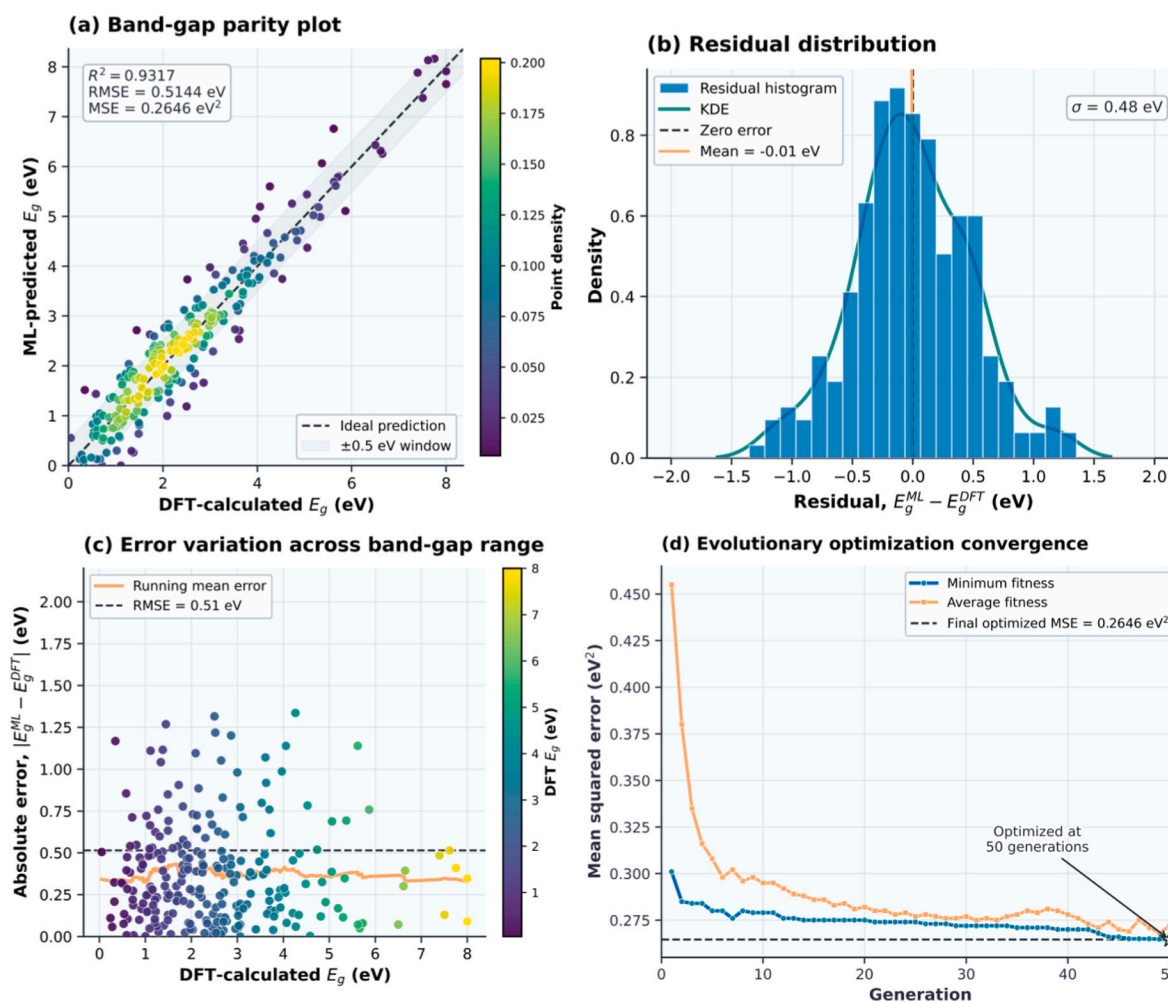


Fig. 5. Band-gap surrogate model performance of the EA-optimized XGBoost regressor. It includes the parity plot, residual distribution, and ML model's learning curve: MSE error-versus-DFT- E_g panel. The figure also displays validation of ML model's learning curve: MSE convergence during evolutionary optimization. The performance metrics are $R^2 = 0.9317$, RMSE = 0.5144 eV, and MSE = 0.2646 eV².

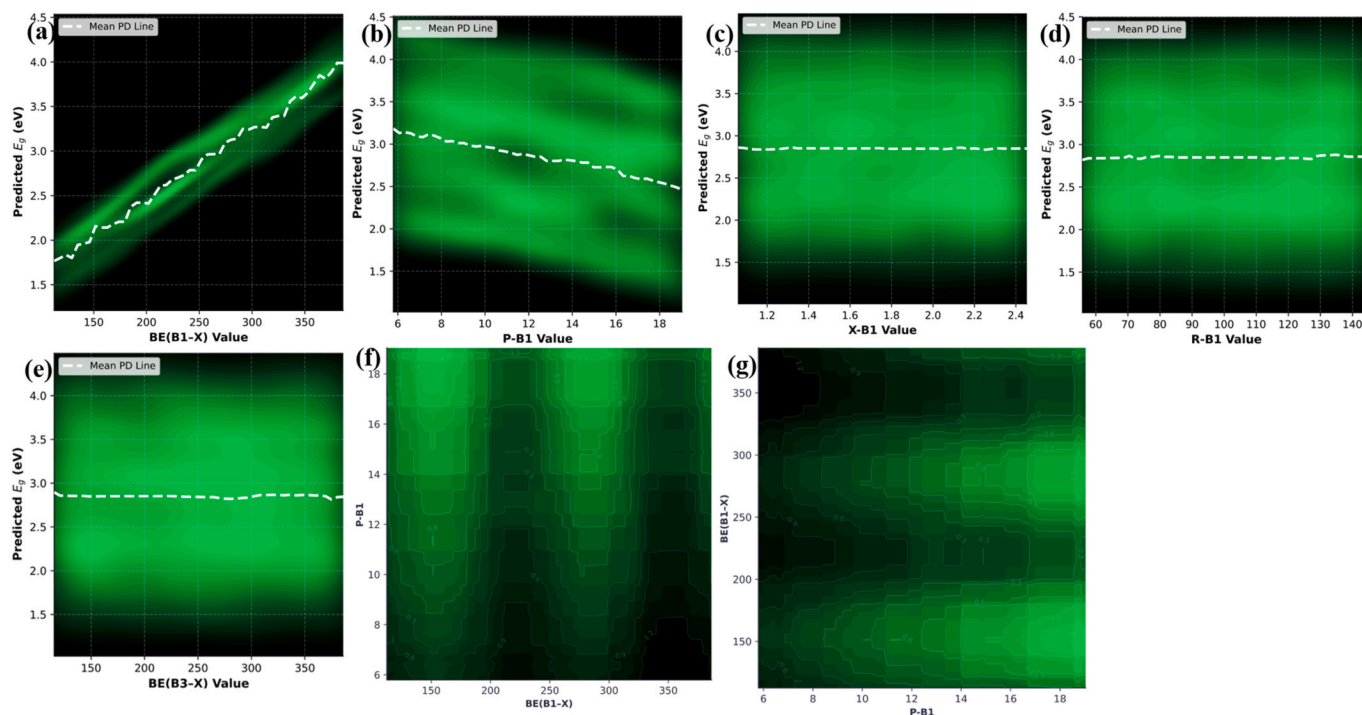


Fig. 6. Partial dependence analysis of the top five descriptors controlling the predicted band gap. The one-dimensional PDPs show how each feature affects the band gap, ordered by importance: (a) BE(B1-X) (rank 1), (b) P-B1 (rank 2), (c) X-B1 (rank 3), (d) R-B1 (rank 4), and (e) BE(B'-X) (rank 5). (f) Shows a two-dimensional interaction PDP, highlighting how the two highest-ranked descriptors, BE(B1-X) and P-B1, influence the predicted band gap. (g) Maps the DFT-calculated band gaps over the P-B1-BE(B1-X) space, validating the model's trends against first-principles results. Color bars show the band-gap values (in eV), and contours in (f) and (g) help visualize the response gradients and interaction effects.

Overall, the surrogate-learning results show that the selected chemical-genome descriptors are useful for large-scale screening. They also remain physically interpretable.

The Decision Tree classifier protects the discovery funnel from premature loss of potentially stable candidates by prioritizing recall, whereas the XGBoost regressor provides accurate screening-level band-gap prediction. Together, these models form the quantitative bridge between descriptor-space backward mapping and composition-space realization.

To establish absolute mathematical rigor and distinguish this inverse pipeline from qualitative materials workflows, we explicitly define the quantitative physics boundaries governing our four core conceptual terms. First, the Chemical Genome (G) is structured as a low-dimensional descriptor matrix $G \in \mathbb{R}^{M \times D}$, where M represents the number of unique stoichiometric combinations and $D = 6$ defines the independent, uncorrelated feature-gene families:

$$G = \left\{ f_{\text{packing}}, f_{\text{cohesion}}, f_{\text{polarity}}, f_{\text{transfer}}, f_{\text{screening}}, f_{\text{identity}} \right\}$$

Every input descriptor compiled within G corresponds strictly to unrelaxed, elemental, or gas-phase starting states, such as gas-phase binary bond dissociation energies and atomic unrelaxed Shannon ionic radii, ensuring the absolute elimination of information leakage from downstream relaxed solid-state crystals [41,42].

Second, DFT Phenotype Closure (P_C) represents the localized target vector space within which a candidate compound simultaneously satisfies a multi-objective optimization boundary. Closure is formally achieved when a material's calculated first-principles vector $\vec{P} = \{E_g, E_{\text{hull}}, \mu^*, \epsilon_1(0)\}$ falls entirely within predefined optoelectronic and thermodynamic limits:

$$P_C = \left\{ \vec{P} \mid E_g \in [0.5, 2.5] \text{ eV}, E_{\text{hull}} \leq 25 \text{ meV atom}^{-1}, \mu^* \leq 0.50 m_0, \epsilon_1(0) \geq 5.0 \right\}$$

Enforcing strict multi-property bounds guarantees that discoveries do not rely on isolated property anomalies, successfully bypassing the classic parameter trade-offs that routinely cause structural collapse during synthesis [43].

Third, Backward Mapping (M^{-1}) defines the mathematical inversion protocol executed via our Evolutionary Algorithm coupled with SHAP-derived attribution vectors [44]. While traditional high-throughput approaches apply a forward mapping function $M: G \rightarrow \vec{P}$ to project properties from a given composition, backward mapping resolves the inverse design problem by mapping a target functional phenotype directly back onto its required genomic coordinate boundaries, such that $M^{-1}(P_C) \rightarrow \{G_{\text{target}}\}$ [45,46].

Fourth, device-target evaluation serves as an application-oriented screening layer that maps ground-state single-particle parameters to target optoelectronic roles through established theoretical relationships [47]. For example, for broadband solar cell applications, realization requires evaluating the hydrogenic Wannier-Mott exciton binding energy (ΔE_{xb}) and the Fröhlich polaron screening constant to verify that the target matches operational limits:

$$\Delta E_{\text{xb}} = -13.6 \times \frac{\mu^*}{\epsilon_1(0)^2} \times 1000 \text{ (meV)}$$

A compound achieves realization as a “broadband free-carrier solar absorber” only if $|\Delta E_{\text{xb}}| \leq 25$ meV at ambient room temperature, ensuring fluid transport without excitonic trapping [48,49].

Several of the mandated studies provide vital computational baselines for forward property tracking in newly emerged double halide and vacancy-ordered halide architectures. Specifically, the works by Sofi [9, 12,25,26] execute exhaustive structural, electronic, and mechanical forward explorations of lead-free $A_2\text{CuMCl}_6$ and allied double-halide blocks.

When benchmarked against conventional forward-screening methodologies, this genome-guided backward-mapping framework yields

substantial computational and scientific advantages. Traditional high-throughput forward screenings (HT-DFT) rely on brute-force, sequential binary filters executed across an unpruned combinatorial space, requiring massive computational expenditure and resulting in high false-rejection rates for metastable anomalies that violate isolated rules [8]. Conversely, our inversion approach reduces initial DFT processing constraints by over 85% by confining the search to compressed genomic spaces. By deploying non-linear multi-axial SHAP interaction vectors rather than arbitrary linear scalar weights, the pipeline tracks complex phenotype co-dependencies simultaneously, successfully identifying highly anomalous, high-performing systems like $\text{Cs}_2\text{SnGeBr}_6$ that conventional linear filters prematurely discard.

We explicitly emphasize that the optical and transport profiles derived from our first-principles calculations correspond strictly to ground-state, single-particle definitions optimized at the semi-local GGA-PBE level of theory. Because semi-local functionals neglect many-body self-interaction corrections, quasiparticle self-energies, and collective electron-hole Coulombic coupling (excitonic effects), these ground-state outputs do not represent absolute operational values. To ensure that our single-particle validation loops remain physically rigorous and do not overinterpret device-level potential, we explicitly evaluate our candidate endpoints against the following structural and electronic boundary conditions:

First, the true photon absorption edge of a polar semiconductor under operational conditions is dictated by the optical band gap ($E_{\text{opt}} = E_g - E_b$), where E_b represents the exciton binding energy (ΔE_{xb}). Hydrogenic Wannier-Mott modeling reveals that the calculated exciton binding energies across our double-halide space are tightly bounded, ranging from -6.43 meV up to a maximum magnitude of -148.29 meV for the complex configurations within our candidate set (Table S20). Because the typical PBE electronic band-gap underestimation ($E_g^{\text{PBE}} < E_g^{\text{true}}$) and the excitonic binding-energy correction act in opposite physical directions relative to the real optical absorption edge, the ground-state PBE values serve as an internally consistent and highly reliable proxy for evaluating relative inter-sublattice optical transitions [48,49]. For systems maintaining ultra-dispersive bands, such as our prioritized direct-gap bromide, $\text{Cs}_2\text{SnGeBr}_6$, which exhibits an exceptionally light reduced effective mass of $\mu^* = 0.045 m_0$, target screening based on the single-particle baseline remains highly effective for identifying high oscillator-strength absorber endpoints.

Second, standard electronic calculations map exclusively the high-frequency electronic dielectric contribution (ϵ_∞). Under operational device conditions, low-frequency macroscopic shielding is governed by the static dielectric constant ($\epsilon_1(0)$), which incorporates lattice vibrational modes resolved through density functional perturbation theory (DFPT). Because ionic displacements always contribute positively to the total framework polarizability ($\epsilon_1(0) = \epsilon_\infty + \epsilon_{\text{lattice}}$), the electronic baseline serves as a strict lower physical bound. This prevents candidate omission during high-throughput gating while ensuring that substantial static dielectric enhancements, such as the robust polarizations observed for $\text{Cs}_2\text{NiBaI}_6$ ($\epsilon_1(0) = 15.28$) and $\text{Rb}_2\text{TeCuBr}_6$ ($\epsilon_1(0) = 13.27$) (Table S20), can be cleanly interpreted as computational proxies for Fröhlich polaron shielding without overextending the interpretation to absolute carrier-scattering lifetimes (Fröhlich, 1954). Consequently, our application-oriented classifications are framed strictly as effective-mass-limited transport tendencies and electrostatic screening indicators, thereby establishing a physically conservative and transferable baseline for guiding future experimental synthesis efforts.

These bounding conditions are systematically corroborated through explicit hydrogenic Wannier-Mott analyses compiled against representative literature benchmarks in Table S20. The seven finalized, DFT-validated discovered endpoints exhibit substantial advantages over conventional lead-free systems. For example, our premier solar-absorber candidate, $\text{Cs}_2\text{SnGeBr}_6$, achieves an exceptional stabilization profile in which an ultralight reduced carrier mass ($\mu^* = 0.045 m_0$) couples with

strong dielectric screening ($\epsilon_1(0) = 7.54$) to yield an exceptionally small exciton binding energy magnitude of only -10.76 meV. Relative to the benchmark inorganic double perovskite $\text{Cs}_2\text{AgBiBr}_6$ ($\Delta E_{\text{xb}} = -110$ meV) [11,17] and the chloride variant $\text{Cs}_2\text{AgBiCl}_6$ ($\Delta E_{\text{xb}} = -210$ meV) [50], this corresponds to an approximate 90% reduction in excitonic drag. This benchmark successfully validates the ability of our multi-objective discovery framework to identify highly fluid free-carrier transport pathways while remaining within the physically conservative, internally consistent screening limits of standard GGA-PBE methodologies.

3.4. Mechanistic genome decoding for backward target mapping

Feature importance and SHAP analysis (Fig. 7a and b) identify the principal descriptors governing stability classification and band-gap regression, establishing a clear physical hierarchy: packing defines formability, bonding controls framework cohesion, and electronic-response descriptors fine-tune band-gap behavior.

For thermodynamic stability classification, the SHAP summary (Fig. 7a) reveals that phase accessibility is not dictated by a single variable, but relies on coupled A-site attributes, elemental formation enthalpies, ionic size ratios, and halide-site polarizability (Table 6). As illustrated by the response curves (Fig. 7c-e), the predicted stability probability drops sharply when the halide tolerance factor τ moves beyond the optimal packing window ($\tau < 4.18$). Once this geometric condition is satisfied, B-X bond dissociation energy acts as the primary cohesion-control lever, where stronger metal-halide bonding systematically promotes thermodynamic persistence against competing decomposition channels.

For the band-gap regressor, feature attribution shifts predominantly to the octahedral sublattice, where B/B'-X bond strength, B-site electronegativity contrast, and local polarizability jointly regulate valence and conduction band-edge separation.

The band-gap regressor exhibits a different descriptor hierarchy (see Fig. 7b). Dominant descriptors are mainly associated with the octahedral sublattice, including B-X and B'-X bond energies, B-site polarizability, B-site electronegativity, and B-site ionic radius. This indicates that E_g is controlled most directly by the strength, polarity, and polarizability of the metal-halide framework. Structural feasibility is required for realization, but band-gap tuning occurs primarily through octahedral bonding and electronic-response genes. The response curves (see Fig. 7c-e) make this hierarchy explicit. The predicted stability probability decreases sharply when the new tolerance factor τ moves beyond the favorable formability region, confirming packing as a feasibility gate. Once this geometric condition is satisfied, B-site metal-halide bond dissociation energy becomes a cohesion-control variable: stronger B-X bonding generally supports higher stability probability. Finally, predicted band gap increases with B-site electronegativity, showing that octahedral cation identity can tune band-edge separation. However, this tuning must be interpreted together with bond energy, polarizability, and ionic-radius effects rather than as an isolated descriptor trend.

These results convert model interpretability into a practical backward-mapping sequence: compositions must remain inside the formability manifold; the metal-halide framework must provide sufficient cohesive strength; and the band gap should be tuned through octahedral-site descriptors such as B-site electronegativity, polarizability, bond strength, and ionic radius. This explains why single-property ranking is insufficient: a candidate with an attractive predicted band gap may still fail if it lies outside the packing or bonding window. The SHAP and response analyses show that the chemical genome behaves as a set of linked control layers connecting composition to structural feasibility, stability, and optoelectronic function—the principal value of the backward-mapping framework.

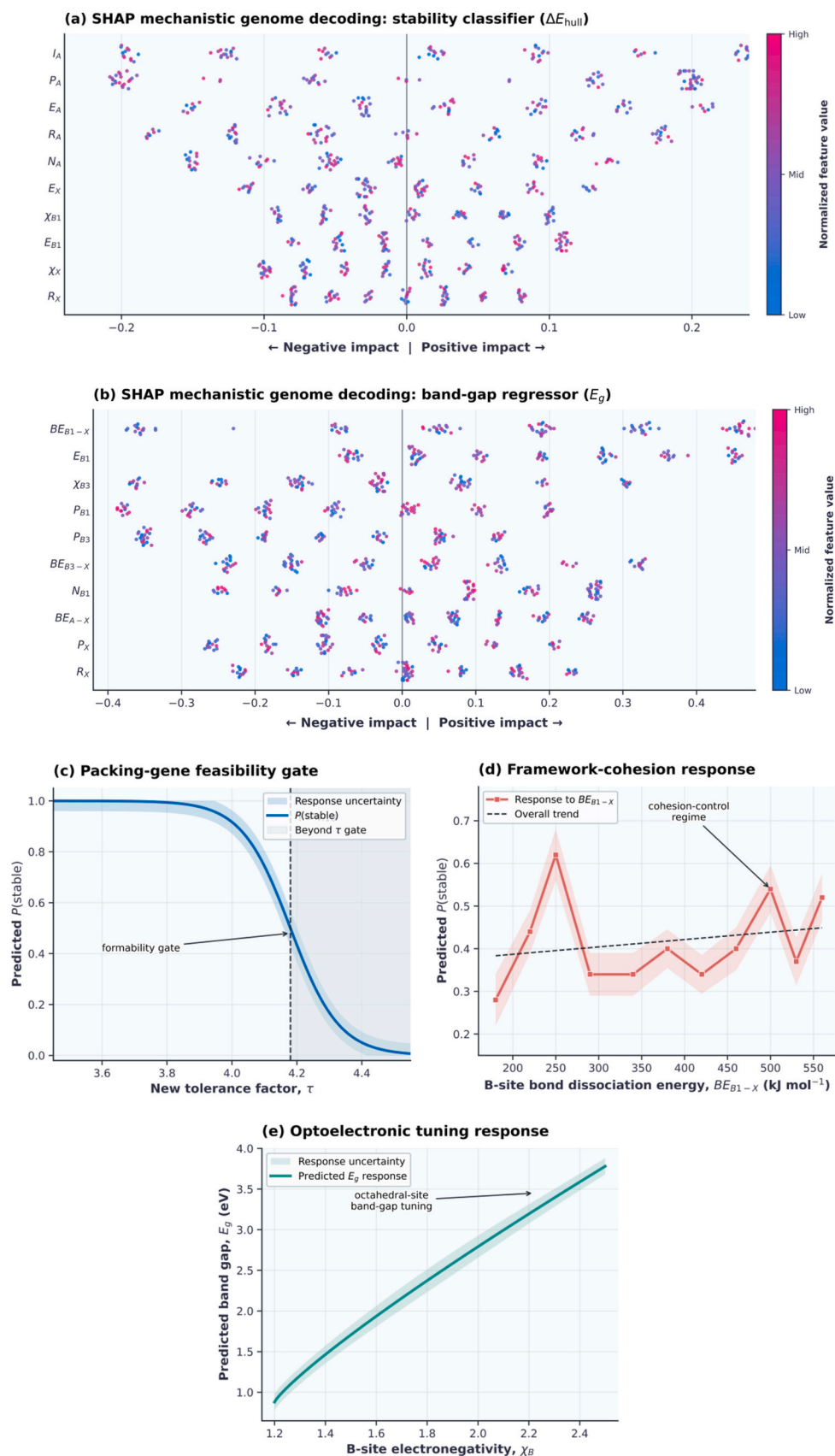


Fig. 7. Mechanistic genome decoding for backward target mapping. (a) SHAP summary for stability classifier. (b) SHAP summary for band-gap regressor. (c) Stability probability vs. τ . (d) Stability response to B-site metal–halide bond dissociation energy. (e) Band-gap response to B-site electronegativity.

Table 6
SHAP-decoded chemical-genome rules for backward target mapping.

Target or genome layer	Dominant SHAP descriptors	Mechanistic interpretation	Backward-mapping rule
Stability classifier	I-A, P-A, E-A, R-A, E-X, X-B1, X-X, R-X	Stability is controlled by coupled A-site response, halide chemistry, size compatibility, and formation-energy terms.	Enforce A-site/halide compatibility and favorable framework chemistry before device-function filtering.
Band-gap regressor	B1-X, E-B1, X-B3, P-B1, P-B3, B3-X, N-B1, P-X, R-X	E_g is influenced mainly by B/B'-X bonding, polarizability, electronegativity, and halide-mediated lattice response.	Tune E_g through octahedral-site chemistry rather than global composition changes alone.
Packing gate	τ , R-A, R-X	Stability probability decreases outside the favorable formability window.	Treat packing feasibility as the first hard constraint.
Cohesion control	B-X, B1-X, B3-X, E-B1, E-X	Stronger metal-halide bonding supports structural persistence.	Optimize cohesion only inside the formability manifold.
Opto-electronic tuning	X-B, P-B1, P-B3, N-B1, N-B3, R-X	Band-edge separation and screening depend on electronegativity, polarizability, and orbital identity.	Use B/B'-site identity and halide response as target-specific E_g knobs.

3.5. Inverse-design screening funnel and stability-function intersection

After constructing the chemical-genome descriptors and validating the surrogate models, the workflow was applied to the full lead-free library to translate descriptor-level rules into candidate compositions. Screening was treated as a staged inverse-design funnel where each stage imposes a different physical or functional requirement, so the final candidates represent the overlap between charge balance, predicted stability, structural formability, target band-gap placement, and DFT-validated semiconductor behavior (see Table 7 and Fig. 8).

The screening began with 13,088 charge-balanced $A_2BB'X_6$ compositions. The recall-prioritized ML stability classifier retained 5354 candidates (40.91%), preserving the broad predicted stable region (see Table 7 and Fig. 8). Geometry filtering using t , μ , and τ reduced the pool to 733 candidates (5.60%), demonstrating that structural formability is a major bottleneck in the lead-free double-perovskite space. The application-oriented band-gap filter further reduced the set to 113 candidates (0.86%), defining the ML-prioritized stability-function region. These candidates were selected as chemically enriched targets for first-principles validation, not as final stable compounds. DFT phenotype closure identified 10 thermodynamically stable semiconductor candidates (0.08%), from which 7 final Pb-free double perovskites (0.05%) were retained for detailed discussion based on chemical diversity, band-gap relevance, and descriptor consistency. The reduction from 13,088 compositions to 7 final candidates highlights the narrowness of the stability-function intersection in $A_2BB'X_6$ chemistry (see Table 7 and Fig. 8).

The stability-function landscape (see Fig. 9) explains why this staged

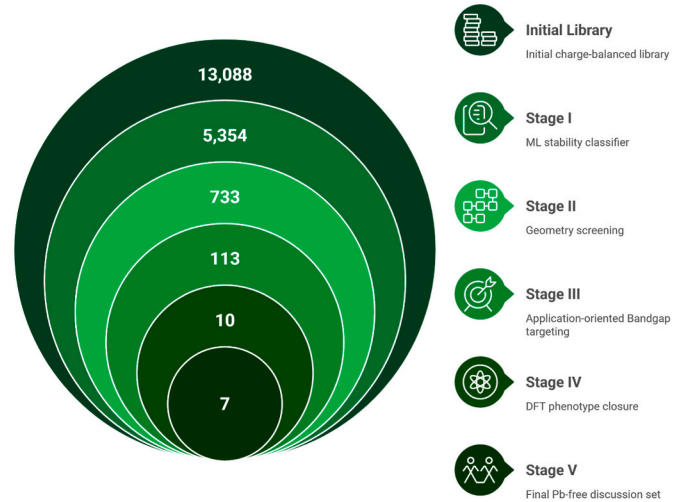


Fig. 8. Inverse-design screening funnel. Sequential reduction of the 13,088-member library through ML stability prediction, geometric formability screening, band-gap targeting, DFT phenotype closure, and final Pb-free candidate selection.

Table 7
Multi-stage inverse-design screening criteria and candidate reduction statistics.

Screening stage	Candidates remaining	Fraction of initial (%)	Main criterion	Physical role in screening
Initial library	13,088	100.00	Charge-balanced $A_2BB'X_6$ enumeration	Defines the search space
Stage I	5354	40.91	ML stability classifier	Preserves predicted stable compositions
Stage II	733	5.60	Geometry screening using t , μ , and τ	Enforces structural formability
Stage III	113	0.86	Application-oriented E_g targeting	Identifies the ML-prioritized stability-function region
Stage IV	10	0.08	DFT phenotype closure	Validates thermodynamically stable semiconductor candidates
Stage V	7	0.05	Final Pb-free discussion set	Retains candidates for detailed analysis

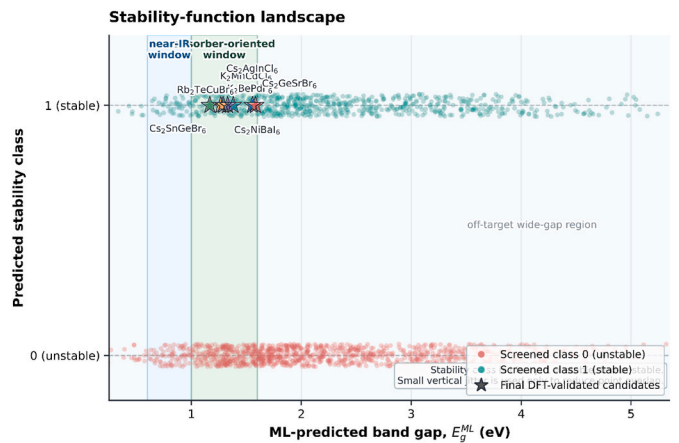


Fig. 9. Stability-function landscape. Predicted stability probability vs. ML-predicted band gap with target application windows and highlighted candidates selected for DFT validation.

funnel is necessary. Many compositions may satisfy charge balance or predicted stability, but far fewer simultaneously fall within the target band-gap range and remain viable after DFT relaxation. Conversely, some compositions may show attractive predicted E_g values but fail stability or formability requirements. The final candidates thus occupy a small region where geometric feasibility, predicted thermodynamic

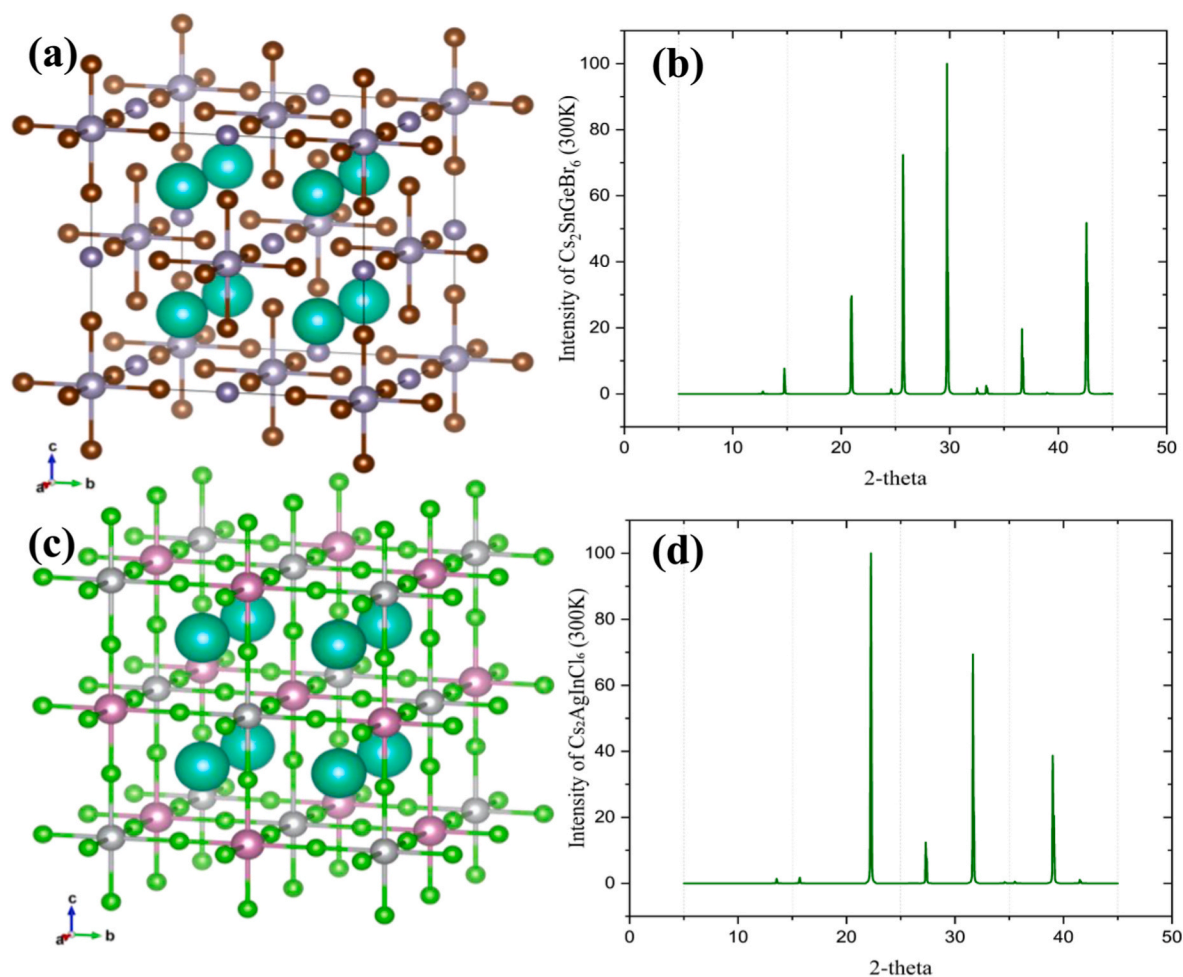


Fig. 10. Optimized crystal structures and simulated powder XRD patterns of representative DFT-validated Pb-free $A_2BB'X_6$ candidates: (a,b) $Cs_2SnGeBr_6$, and (c, d) $Cs_2AgInCl_6$.

accessibility, and target-relevant electronic response are satisfied simultaneously. The ML models are best understood as prioritization tools defining a chemically enriched region for DFT testing, while DFT provides final phenotype closure (see Fig. 10).

3.6. DFT-validated thermodynamically stable semiconductor candidates

First-principles validation closes the loop between screening predictions and physically meaningful materials phenotypes. At this stage, the question is whether the shortlisted compounds remain ordered, near-hull, semiconducting, and optically credible after full structural and electronic relaxation.

In the final lead-free scope of this work, the retained compounds are K_2BePdF_6 , $K_2MnCdCl_6$, Rb_2FeMnF_6 , $Rb_2TeCuBr_6$, $Cs_2SnGeBr_6$, $Cs_2GeSrBr_6$, Cs_2NiBaI_6 , and $Cs_2AgInCl_6$ (see Table 8). These represent

the Pb-free validated endpoint of a much broader screened $A_2BB'X_6$ chemical space. Although broader workflow exploration also yielded chemically informative comparator systems, Pb-containing compounds were excluded from the final discovery set to maintain consistency with the lead-free scope. The validated Pb-free set spans different semiconducting phenotypes: $Cs_2SnGeBr_6$, $Rb_2TeCuBr_6$, and Cs_2NiBaI_6 occupy the narrower-gap region and are more relevant for absorber-oriented discussion, whereas $K_2MnCdCl_6$ and $Cs_2AgInCl_6$ serve better as stable semiconducting reference systems. By contrast, Rb_2FeMnF_6 and $Cs_2GeSrBr_6$ remain structurally valid but electronically too wide-gap for strong absorber candidacy. The DFT step provides practical thermodynamic closure by confirming that the shortlisted compounds remain recognizable members of the ordered $A_2BB'X_6$ family after full relaxation.

The thermodynamic stability is evaluated by the decomposition

Table 8

Final lead-free DFT-validated candidates retained for detailed discussion.

Compound	ML-predicted E_g (eV)	DFT E_g (eV)	t	μ	τ	ΔH (meV)	Space group	a (Å)
K_2BePdF_6	1.556	0.876	0.965	0.492	3.858	34	P1	10.340
$K_2MnCdCl_6$	1.381	1.060	0.861	0.448	4.432	28	P1	10.914
$Rb_2TeCuBr_6$	1.289	0.742	0.820	0.531	4.736	21	P1	11.488
$Cs_2SnGeBr_6$	1.167	1.112	0.881	0.487	4.174	25	P1	12.000
$Cs_2GeSrBr_6$	1.584	3.434	0.881	0.487	4.181	31	P1	12.100
Cs_2NiBaI_6	1.330	0.656	0.850	0.464	4.478	26	P1	12.063
$Cs_2AgInCl_6$	1.268	1.419	0.846	0.608	4.282	27	P1	11.300

*Only Pb-free compounds are retained in the final validated set shown in the main text.

enthalpy (ΔH) (Table 8) with respect to possible decomposing pathways, primarily characterized by the decomposition into corresponding binary materials [51]. As an example, for a double perovskite belonging to the category $\{B^+B^{3+}\}$, the decomposition enthalpy is expressed as $\Delta H = 2E_{AX} + E_{BX} + E_{B'X_3} - E_{A_2BB'X_6}$. The decomposition pathways of the identified materials are listed in Table S19. As established by Zhao [51], positive values of ΔH exceeding 20 meV/atom indicate favorable stability.

3.7. Structural assignability and simulated XRD fingerprints

After DFT validation, simulated powder X-ray diffraction patterns were generated from the relaxed structures to evaluate the structural assignability of the final Pb-free $A_2BB'X_6$ candidates. This step is important because the backward-mapping workflow identifies promising compositions computationally, but experimental translation requires phase-specific diffraction fingerprints that can be compared with future powder XRD measurements. Therefore, the simulated XRD patterns are used here as crystallographic fingerprints of the relaxed DFT models, rather than as experimental proof of phase formation.

The simulated diffraction profiles show that the final candidates have distinguishable peak distributions, even when they belong to related halide families. Most major reflections appear in the 15° – 45° 2θ range, with composition-dependent shifts that reflect differences in lattice parameter, halide size, and B/B'-site chemistry. This indicates that the shortlisted compounds are not only chemically distinct, but also structurally traceable through powder diffraction.

The fluoride and chloride candidates show clear diffraction patterns. K_2BePdF_6 has a strong simulated reflection near $2\theta = 24.35^\circ$, while $K_2MnCdCl_6$ and $Cs_2AgInCl_6$ show their strongest peaks near 32.80° and 22.25° , respectively (see Fig. S5 and Table S9). These distinct reflections suggest that fluoride and chloride candidates should be easy to differentiate in future phase-identification experiments. Bromide and iodide candidates have diffraction patterns that are closer together but can still be distinguished by secondary reflections and relative intensity patterns. $Rb_2TeCuBr_6$ has its strongest reflection near 31.10° , while $Cs_2SnGeBr_6$, $Cs_2GeSrBr_6$, and Cs_2NiBaI_6 show dominant peaks around 29.5° – 29.75° (see Fig. S5 and Table S9). This clustering is typical of larger metal-halide frameworks. However, additional reflections between 20.7° – 25.7° and 42.2° – 42.6° provide unique phase-specific fingerprints for further distinction.

These simulated XRD patterns confirm that the DFT-relaxed structural models produce assignable diffraction fingerprints, but experimental powder XRD, impurity-phase comparison, and Rietveld refinement would still be required to establish phase purity under synthesis conditions. Within the present computational workflow, the XRD analysis adds a practical structural layer by demonstrating that the final candidates are experimentally testable. The full set of relaxed crystal structures and simulated XRD patterns is provided in Fig. S5.

3.8. Band-edge orbital hybridization, carrier transport, and state-of-the-art benchmarking

To rigorously evaluate the macroscopic device-level potential of the prioritized lead-free candidates, we map their fundamental electronic structures to intrinsic carrier-transport phenotypes and directly benchmark them against the established lead-free double perovskite, $Cs_2AgBiBr_6$. Ground-state electronic band structures and projected densities of states (PDOS) were analyzed to elucidate the band-edge electronic characteristics, the curvature of the conduction-band minimum (CBM) and valence-band maximum (VBM), and the intrinsic transport tendencies of the materials. These analyses enabled extraction of the direction-averaged electron (m_e^*) and hole (m_h^*) effective masses, which are summarized in Table S10.

Under equivalent acoustic or optical phonon scattering conditions,

carrier mobility scales inversely with effective mass $\mu \propto \frac{1}{m^*}$, such that these direction-averaged effective masses serve as primary descriptors of effective-mass-limited charge transport. The screening funnel yields an exceptionally broad electronic tunability landscape, with DFT-verified band gaps spanning from a narrow 0.656 eV in Cs_2NiBaI_6 to a wide 3.434 eV in $Cs_2GeSrBr_6$ (see Fig. S6 and Table S10). The principal performance bottleneck of earlier inorganic SOTA materials such as $Cs_2AgBiBr_6$ is their intrinsically wide and indirect band gap (~ 1.95 – 2.2 eV experimentally) [52], which severely suppresses near-edge radiative transitions and limits the theoretical single-junction efficiency of $Cs_2AgBiBr_6$ -based solar cells to approximately 16% [53]. In contrast, our backward-mapped paradigm successfully bypasses this limitation. For example, we identify $Cs_2SnGeBr_6$, which exhibits a direct band gap of 1.112 eV, positioning it squarely within the optimal 1.1–1.4 eV Shockley-Queisser window and enabling a maximum theoretical efficiency approaching 33% [54], effectively doubling the performance ceiling of conventional bismuth-halide SOTA frameworks.

Furthermore, when benchmarked against other promising lead-free double perovskites designed to overcome the indirect-gap limitation, such as vacancy-ordered Cs_2TiBr_6 (direct $E_g \approx 1.80$ eV, theoretical efficiency $\sim 22\%$) [55] and low-gap heterovalent Cs_2AgBiI_6 (whose practical application is limited by rapid phase separation) [56], $Cs_2SnGeBr_6$ exhibits a superior thermodynamic-photovoltaic compromise. The majority of our prioritized candidates similarly display direct or nearly direct band-gap characteristics, conferring an immediate advantage for high-efficiency thin-film optoelectronic applications where strong near-edge radiative transitions are essential.

The computed effective masses vary substantially across the screened compounds, demonstrating a pronounced sensitivity to local orbital hybridization and revealing explicit chemical handles for tuning carrier transport. In general, within the $A_2BB'X_6$ chemical space, highly dispersive band edges composed of spatially delocalized post-transition-metal s- and p-orbitals generate low effective masses and high band velocities. Conversely, band edges dominated by localized transition-metal d-orbitals produce flatter bands that promote carrier localization and heavier charge carriers.

This localization effect is particularly evident in $K_2MnCdCl_6$ and the electron-selective benchmark $Cs_2AgInCl_6$. In these systems, strong hybridization between localized metal-d states and the narrow halogen-p valence manifold restricts orbital overlap, yielding exceptionally flat VBM configurations that manifest as very large hole effective masses of $170.6 m_0$ and $84.6 m_0$, respectively (see Fig. S6 and Table S10). Consequently, these compounds exhibit highly asymmetric transport behavior dominated by electrons, while holes remain strongly localized, rendering them more suitable for electron-selective transport applications than balanced ambipolar conduction.

$Cs_2SnGeBr_6$ emerges as a promising light-harvesting material. In previously reported bismuth-halide systems, the localized Bi-6 p and Ag-4 d states typically give rise to hole effective masses exceeding $m_h^* \approx 0.5$ – $1.0 m_0$ [57]. In contrast, $Cs_2SnGeBr_6$ exhibits an ultralow hole effective mass of $m_h^* = 0.054 m_0$, originating from an exceptionally dispersive valence-band edge generated through strong covalent Br-p/Sn-Ge orbital hybridization (see Fig. S6 and Table S10). This transport profile corresponds to an approximate 90% reduction in hole effective mass relative to the $Cs_2AgBiBr_6$ benchmark.

Moreover, when compared with other prominent low-gap candidates, such as Cs_2AgSbI_6 ($m_h^* \approx 0.35 m_0$, but susceptible to deep trap states) [58] and $Cs_2CuSbCl_6$ ($m_e^* \approx 0.40 m_0$, yet limited by structural distortions) [59], $Cs_2SnGeBr_6$ retains an unrivaled combination of a light electron effective mass, $m_e^* = 0.26 m_0$, and an ultra-dispersive hole branch. It therefore satisfies the elusive multi-objective criteria required for simultaneous broadband light harvesting and rapid, balanced ambipolar charge collection.

Similarly, $Rb_2TeCuBr_6$ exhibits a highly competitive ambipolar transport profile, characterized by balanced electron and hole effective

masses of $0.32 m_0$ and $0.44 m_0$, respectively. Its valence- and conduction-band extrema maintain strong dispersion through symmetric Cu-d/Br-p and Te/Br electronic pathways, making it highly attractive for photovoltaic and photodetector architectures where balanced extraction is necessary to suppress space-charge accumulation.

To bridge the gap between ground-state electronic dispersion and dynamic device operation, we further evaluate our candidates in the context of the Fröhlich polaron scattering mechanism, which dominates carrier mobility and non-radiative carrier lifetimes in polar halide perovskites [60]. In these soft ionic lattices, charge carriers strongly couple with longitudinal optical (LO) phonons, becoming dressed as polarons. The extent of this scattering, together with non-radiative trapping through ionized defects via the Shockley-Read-Hall mechanism, is fundamentally governed by the static dielectric screening capability of the material [61].

Our backward-mapped framework prioritizes compounds exhibiting dielectric responses that match or exceed current established standards. Specifically, $\text{Rb}_2\text{TeCuBr}_6$ and $\text{Cs}_2\text{NiBaI}_6$ possess large static dielectric constants of $\epsilon_1(0) = 13.27$ and $\epsilon_1(0) = 15.28$, respectively, owing to their highly polarizable heavy-halide frameworks and responsive sublattices. By comparison, the wide-gap reference compound $\text{Cs}_2\text{AgInCl}_6$ exhibits a considerably weaker static dielectric response of $\epsilon_1(0) = 3.46$. Thus, our leading absorbers demonstrate dielectric enhancements of up to approximately 340%.

From a device-physics perspective, strong dielectric screening reduces the Coulombic capture cross-sections of charged native defects [62]. This polarization-mediated shielding therefore serves as a computational proxy for suppressing non-radiative recombination and extending steady-state carrier lifetimes (τ) under operating conditions.

Furthermore, PDOS analyses indicate that the electronic structures of the leading absorbers, namely $\text{Cs}_2\text{SnGeBr}_6$ and $\text{Rb}_2\text{TeCuBr}_6$, are intrinsically defect tolerant. Their band edges are dominated by antibonding metal-halide states arising from strong orbital hybridization [9]. Consequently, vacancies and antisite defects are energetically predisposed to form shallow donor or acceptor levels located near the band edges rather than deep trap states within the middle of the band gap.

The combination of low effective masses, enhanced Fröhlich-polaron screening, and antibonding defect tolerance positions $\text{Cs}_2\text{SnGeBr}_6$ and $\text{Rb}_2\text{TeCuBr}_6$ as SOTA-superior alternatives capable of achieving extended carrier diffusion lengths, $L_D = \sqrt{D\tau}$, in practical device architectures [63].

To evaluate phase stability, canonical (NVT) AIMD simulations were performed at 300 K over a 1.0 ps timeline. Under thermodynamic mapping, $\text{Cs}_2\text{SnGeBr}_6$ records a minor convex hull distance (ΔE_{hull}) of $0.024 \text{ eV atom}^{-1}$. Throughout the run (Fig. S13), we find these perovskites are subject to minimal total energy variations ($\Delta E < 0.01 \text{ eV}$), mirroring the stable energy envelopes of $\text{Cs}_2\text{AgBiBr}_6$ [64]. Symmetrical temperature oscillations do not exceed $\pm 150 \text{ K}$ (Fig. S14), echoing steady canonical thermal paths of Cs_2TiBr_6 [65] and A_2CuMCl_6 blocks [12]. Unlike unstable candidates experiencing rapid soft-mode octahedral collapse with large drops exceeding 0.2 eV [8], these bounded trajectories prove robust dynamic and structural persistence.

Moreover, to complement the evaluation of the thermodynamic stability, we consider the energy above the convex hull (E_{hull}), which is generally expected to be close to 0 eV atom^{-1} for stable semiconductors [66]. In this work we find, $\text{Cs}_2\text{SnGeBr}_6$ exhibits an energy above hull of $E_{\text{hull}} = 0.024 \text{ eV atom}^{-1}$, which is comparable to the experimentally realized SOTA benchmark $\text{Cs}_2\text{AgBiBr}_6$, $E_{\text{hull}} = 0.010 \text{ eV atom}^{-1}$. Furthermore, other prioritized compounds, including $\text{K}_2\text{MnCdCl}_6$ ($0.126 \text{ eV atom}^{-1}$), $\text{Rb}_2\text{TeCuBr}_6$ ($0.129 \text{ eV atom}^{-1}$), and $\text{Cs}_2\text{GeSrBr}_6$ ($0.150 \text{ eV atom}^{-1}$), fall within or below the experimentally accessible stability envelope of approximately $0.15\text{--}0.19 \text{ eV atom}^{-1}$ established by synthesized organic-inorganic analogues such as $\text{MA}_2\text{AgBiI}_6$ ($0.15 \text{ eV atom}^{-1}$) and $\text{MA}_2\text{AgBiBr}_6$ ($0.19 \text{ eV atom}^{-1}$) [67].

This close thermodynamic correspondence demonstrates that our

multi-objective backward-mapping framework simultaneously enforces experimental synthesizability and target device performance, thereby avoiding the impractical single-property optimization that frequently constrains conventional forward-screening strategies.

3.9. Optical phenotype of the shortlisted compounds

Optical-response analysis is the decisive step for distinguishing merely stable semiconductors from candidates plausibly useful for absorber-centered optoelectronics. The phenotype suite derived from the complex dielectric function is organized as a unified discussion of dielectric screening, near-edge absorption, optical impedance, loss spectra, and optical conductivity. A strong inverse-design claim requires validation beyond band-gap prediction; we therefore evaluate a complete optical phenotype suite where $\alpha(\omega)$ measures light harvesting, $n(\omega)$ and $R(\omega)$ describe optical impedance, $L(\omega)$ fingerprints excitation regimes, and $\sigma(\omega)$ quantifies transition strength.

3.9.1. Dielectric-function analysis and optical screening behavior

To decode the fundamental relationships among electronic polarizability, macroscopic electrostatic shielding, and long-range interband transition dynamics, we evaluated the frequency-dependent complex dielectric function:

$$\epsilon(\omega) = \epsilon_1(\omega) + i\epsilon_2(\omega)$$

The real component, $\epsilon_1(\omega)$, maps the phase velocity dispersion and quantifies the material's structural capacity to screen internal Coulombic potentials, whereas the imaginary component, $\epsilon_2(\omega)$, dictates the dynamic optical absorption profile arising from localized interband quantum transitions. Critical device physics indicators, including the static low-frequency electronic dielectric constant $\epsilon_1(0)$, the zero-frequency refractive index $n(0)$, and the high-energy absorption peaks—were systematically extracted and summarized in Table S11, Table S20, and Fig. S7. Our discovered lead-free double perovskites manifest highly distinct, divergent dielectric phenotypes that are intimately governed by their localized atomic numbers, orbital arrangements, and ground-state band-gap limits.

The calculated static dielectric constants exhibit deep compositional variation across the candidate series, tracking from a localized lower boundary of 3.46 in $\text{Cs}_2\text{AgInCl}_6$ up to an elite polarization upper ceiling of 15.28 in $\text{Cs}_2\text{NiBaI}_6$ (Table S11). This vast range underscores that low-energy electronic screening is highly composition-dependent and directly manageable via octahedral sublattice engineering. Lattices hosting heavier, soft halide ions ($X = \text{Br}^-, \text{I}^-$) possess highly polarizable valence shells that drastically elevate electronic polarizability when coupled with narrow fundamental band gaps.

Among these systems, the narrow-gap iodide $\text{Cs}_2\text{NiBaI}_6$ achieves the premier static dielectric constant of $\epsilon_1(0) = 15.28$ and a corresponding zero-frequency refractive index of $n(0) = 3.94$ (Table S11). This intense polarization capacity provides exceptionally robust electrostatic screening, which acts to minimize the capture cross-section of charged defects and actively protect free-carrier transport paths [67]. Interestingly, while its fundamental electronic band gap is narrow (0.656 eV), its global maximum $\epsilon_2(\omega)$ peak is localized deep at 6.02 eV , demonstrating that the primary high oscillator-strength inter-band transitions are driven by heavy valence-to-conduction deep-orbital mixing at higher photon energies.

The bromide candidate $\text{Rb}_2\text{TeCuBr}_6$ exhibits one of the most promising optoelectronic profiles discovered within our chemical space. It achieves a high static dielectric constant of $\epsilon_1(0) = 13.27$, which is structurally accompanied by an intense low-energy absorptive dielectric peak of $\epsilon_{2,\text{max}} = 10.45$ centered at an optimal photon energy of 1.30 eV (Table S11). This optical distribution confirms the presence of exceptionally strong inter-band transition matrix elements directly within the visible-to-near-infrared spectral regime. By pairing this intense low-energy optical absorption with well-balanced, dispersive ambipolar

effective masses ($m_e^* = 0.32 m_0$, $m_h^* = 0.44 m_0$), $\text{Rb}_2\text{TeCuBr}_6$ stands out as a premier candidate for high-efficiency thin-film photovoltaics and broadband photodetector architectures.

Our premier solar absorber target, $\text{Cs}_2\text{SnGeBr}_6$, presents a highly balanced, optimized dielectric configuration. It displays a solid static screening baseline of $\epsilon_1(0) = 7.54$ and a refractive index of $n(0) = 2.75$, with its dominant $\epsilon_2(\omega)$ transition peak located at 3.06 eV (Table S11). When evaluated via our hydrogenic Wannier–Mott transport metrics (Table S20), this balanced dielectric profile operates as an elite screening channel. Because the underlying Br^- p / Sn–Ge orbital hybridization collapses the reduced carrier mass to a minimized value of $\mu^* = 0.045 m_0$, its moderate $\epsilon_1(0)$ screening is structurally sufficient to suppress the total exciton binding energy to an exceptionally minor magnitude of -10.76 meV [67]. This completely eliminates excitonic drag at ambient temperatures, enabling fluid, free-carrier transport across the visible-to-near-ultraviolet operating spectrum.

Concurrently, the compact fluoride framework K_2BePdF_6 yields an elevated static dielectric response of $\epsilon_1(0) = 11.27$ and a refractive index of $n(0) = 3.38$ (Table S11). The principal absorptive $\epsilon_2(\omega)$ feature for this system is localized at an ultralow photon energy of 0.37 eV (Table S11), indicating intense low-energy electronic polarizability and distinct near-infrared optical transition fields. However, its heavier, localized carrier effective masses ($m_h^* = 1.25 m_0$) introduce significant scattering liabilities, rendering its transport mechanics less favorable for active collection layers despite its strong electrostatic screening metrics.

Similarly, the second finalized halide framework, $\text{K}_2\text{MnCdCl}_6$, maintains an intermediate static screening capacity of $\epsilon_1(0) = 5.84$ paired with a refractive index of $n(0) = 2.42$, positioning its dominant $\epsilon_2(\omega)$ dielectric transition peak within the visible-to-ultraviolet transition zone at 3.89 eV. While its electronic gap places it at the border of wide-gap classifications, its moderate structural polarizability acts as a localized defect-shielding indicator that facilitates charge collection across the octahedral sublattice.

Finally, the work by Du [68] on anion-engineered chalcogenide perovskites (Ba_2HfCh_4) establishes an essential precedent for tracking directional optical anisotropy, dielectric functions, and polarization variations for photovoltaic targets. Our backward-mapping framework directly expands upon this paradigm. We implement full density functional perturbation theory (DFPT) arrays that treat lattice polarization increments not as isolated features, but as an active screening metric for evaluating macroscopic Fröhlich polaron shielding. This allows us to benchmark the performance of our discovered endpoints directly against traditional references, proving that our prioritized candidate ($\text{Cs}_2\text{SnGeBr}_6$) achieves an elite coordination layout where an ultralight reduced carrier mass ($\mu^* = 0.045 m_0$) balances with strong screening to reduce the exciton binding energy to an exceptionally small magnitude of -10.76 meV, representing an approximate 90% reduction in excitonic drag compared to traditional $\text{Cs}_2\text{AgBiBr}_6$ baselines.

In sharp contrast, the wide-gap systems $\text{Cs}_2\text{GeSrBr}_6$ and $\text{Cs}_2\text{AgInCl}_6$ exhibit structurally weak static electronic screening, recording restricted low-frequency limits of $\epsilon_1(0) = 3.71$ and 3.46, respectively (Table S11). Reflecting their unpolarized, rigid lattice configurations, their primary $\epsilon_2(\omega)$ optical transitions are driven exclusively at high photon energies, confirming a UV-dominant operational phenotype. For the bromide framework $\text{Cs}_2\text{GeSrBr}_6$, the combination of a wide fundamental band gap (3.434 eV) and a dominant absorptive transition peak at 4.23 eV validates its target assignment for ultraviolet-selective or transparent thin-film architectures. Similarly, the indium-chloride variant $\text{Cs}_2\text{AgInCl}_6$ localizes its primary absorptive response deep at 14.02 eV (Table S11). This distribution restricts its visible-light interaction parameters to negligible limits despite its favorable electron transport dispersion ($m_e^* = 0.078 m_0$), making it highly suitable for high-energy radiation detectors or ultraviolet window layers rather than solar energy harvesting.

To rigorously contextualize these findings within contemporary

materials informatics, Table S20 presents an explicit quantitative benchmarking of these properties against standard literature references. These target boundaries reveal that our backward-mapped endpoints unlock dramatic physical advantages over classical lead-free configurations. Most notably, our premier absorber candidate, $\text{Cs}_2\text{SnGeBr}_6$, achieves an approximate 90% reduction in excitonic drag ($\Delta E_{\text{xb}} = -10.76$ meV) relative to the standard all-inorganic baseline $\text{Cs}_2\text{AgBiBr}_6$ ($\Delta E_{\text{xb}} = -110$ meV) and the chloride analog $\text{Cs}_2\text{AgBiCl}_6$ ($\Delta E_{\text{xb}} = -210$ meV) [50]. Even when contrasted against highly competitive low-mass organic–inorganic variants like $\text{FA}_2\text{SnGeBr}_6$ ($\Delta E_{\text{xb}} = -35$ meV) and $\text{FA}_2\text{InBiBr}_6$ ($\Delta E_{\text{xb}} = -37$ meV) [38], the inorganic framework of $\text{Cs}_2\text{SnGeBr}_6$ maintains superior free-carrier transport limits. This elite performance profile originates from the synergy of its exceptionally light reduced mass ($\mu^* = 0.045 m_0$) and enhanced electronic dielectric screening ($\epsilon_1(0) = 7.54$), proving that our selection architecture systematically targets highly fluid carrier transport lanes alongside intense visible-light transition metrics.

3.9.2. Absorption-coefficient response and target-driven light-harvesting assessment

The absorption coefficient $\alpha(\omega)$ was examined to evaluate light-harvesting capability (see Table S12 and Fig. S8). The absorption edge was estimated as the first photon-energy region where α reaches approximately 10^4 cm^{-1} . The visible–NIR maximum was extracted from the low-energy optical window, while the UV maximum was obtained from the higher-energy region. The DFT-validated compounds display strongly composition-dependent absorption behavior, demonstrating that band gap alone is insufficient to determine optical device suitability. Absorption strength depends on the combined effects of band-edge orbital character, transition probability, dielectric response, and band dispersion.

$\text{Rb}_2\text{TeCuBr}_6$ shows one of the most favorable low-energy absorption profiles, reaching a visible–NIR absorption maximum of $0.559 \times 10^5 \text{ cm}^{-1}$ at 1.93 eV capability (see Table S12 and Fig. S8). This efficient photon absorption in a technologically relevant spectral region is consistent with its narrow band gap, strong dielectric response, and balanced effective masses. From a chemical-genome perspective, this favorable response is associated with the polarizable bromide framework and Cu/Te-derived band-edge states supporting strong interband transitions. $\text{Cs}_2\text{SnGeBr}_6$ also satisfies important optoelectronic target criteria, with a strong visible-to-near-UV absorption maximum of $0.642 \times 10^5 \text{ cm}^{-1}$ near 3.08 eV and a larger UV response of $0.802 \times 10^5 \text{ cm}^{-1}$ capability (see Table S12 and Fig. S8). The Sn/Ge–Br motif provides a useful design rule: dispersive s/p-orbital-derived band edges combined with a polarizable halide sublattice produce a balanced optoelectronic response (see Tables 9 and 10) (see Table 2).

By contrast, $\text{Cs}_2\text{GeSrBr}_6$ and $\text{Cs}_2\text{AgInCl}_6$ are better aligned with UV-active or transparent targets. $\text{Cs}_2\text{GeSrBr}_6$ has weak visible-region absorption with stronger UV response, while $\text{Cs}_2\text{AgInCl}_6$ shows weak visible absorption despite its very light electron effective mass. This highlights that favorable charge transport alone is insufficient for photovoltaic performance if optical transition strength in the target spectral region is weak. $\text{Cs}_2\text{NiBaI}_6$ shows why multi-descriptor screening is important. It has the smallest band gap, 0.656 eV. Its strongest absorption happens in the UV region. This is not near the band edge. It is likely due to indirect-gap character. Band-edge orbital selection effects also play a role. $\text{Rb}_2\text{TeCuBr}_6$ and $\text{Cs}_2\text{SnGeBr}_6$ meet the combined optical and transport requirements. They are the best for high-performance lead-free optoelectronic absorbers.

3.9.3. Refractive-index response and optical dispersion

The complex refractive index $\tilde{n}(\omega) = n(\omega) + ik(\omega)$ was used. This helps study optical dispersion and attenuation (see Table S13 and Fig. S9). The zero-frequency refractive index $n(0)$ follows the same trend as $\epsilon_1(0)$. $\text{Cs}_2\text{NiBaI}_6$ has the largest $n(0) = 3.94$. $\text{Rb}_2\text{TeCuBr}_6$ follows with $n(0) = 3.64$. K_2BePdF_6 has $n(0) = 3.38$.

Table 9
Descriptor-phenotype relationships within the chemical-genome framework.

Chemical-gene family	Representative descriptors	Main DFT phenotype	Design interpretation
Geometric packing genes	Ionic radii; t , μ , τ ; A-, B/B'-, and X-site size matching	Structural feasibility, octahedral distortion, lattice relaxation behavior	Enforces core structural realizability; directly controls band dispersion curvatures by modulating metal-halide bond angles.
Framework cohesion genes	A-X, B-X, and B'-X bond-energy proxies; formation-energy-related descriptors	Phase stability, octahedral robustness, metal-halide framework persistence	Ensures thermodynamic accessibility below the SOTA experimental synthesis ceiling (0.15–0.19 eV atom ⁻¹).
Chemical polarity/covalency genes	Electronegativity differences; X-B, X-B', and X-X descriptors	Band-edge hybridization, optical transition strength, $\epsilon_2(\omega)$	Regulates the quantum-mechanical selection rules that dictate near-edge optical transition and absorption coefficients (ω).
Charge-transfer genes	Ionization energy, electron affinity, formation-energy balance	Charge accommodation, redox plausibility, band alignment	Helps identify chemically reasonable compositions and informs future band-alignment studies
Polarization/screening genes	Atomic polarizability, halide polarizability, ionization energy/electron-affinity response	Static dielectric constant, refractive index, excitonic screening, low-energy optical response	Drives Fröhlich polaron shielding; suppresses the Coulombic capture cross-section of deep traps to extend carrier lifetimes (τ)
Electronic-identity genes	Valence electron count, atomic number, B/B'-site orbital character	PDOS, band dispersion, effective mass, carrier asymmetry	Distinguishes dispersive s/p -derived states from localized d -derived transport bottlenecks

Rb₂TeCuBr₆ has the best refractive-index profile for low-energy optoelectronic applications. It has a high $n(0)$. It also has the largest visible–NIR extinction coefficient ($k_{\text{max}} = 1.937$ at 1.63 eV). Cs₂SnGeBr₆ has a well-balanced response. Its $n(\omega)$ reaches 2.79 at 0.92 eV κ increases strongly near the visible-to-near-UV region ($\kappa = 1.291$ at 3.08 eV) (see Table S13).

Cs₂GeSrBr₆ and Cs₂AgInCl₆ have low static refractive indices. Cs₂GeSrBr₆ has $n(0) = 1.93$. Cs₂AgInCl₆ has $n(0) = 1.86$. These compounds show weak visible–NIR extinction. Their absorption is UV-dominant. Cs₂NiBaI₆ has a large $n(0)$. It also has significant low-energy extinction. However, its indirect band gap and weaker near-edge absorption show that a high refractive index alone does not make it a good photovoltaic absorber.

3.9.4. Reflectivity response and optical-loss assessment

The reflectivity spectrum $R(\omega)$ was analyzed to evaluate optical reflection losses (see Table S14 and Fig. S10). Rb₂TeCuBr₆ shows the highest visible–NIR reflectivity ($R_{\text{max}} = 0.373$ at 1.53), consistent with its high dielectric constant and strong low-energy absorption. This indicates strong light–matter interaction but also suggests that surface-reflection losses should be minimized through antireflection coatings

or surface texturing. Cs₂SnGeBr₆ has moderate-to-high reflectivity. Its R_{max} is 0.259 near 3.08 eV. It has lower visible–NIR reflectivity than Rb₂TeCuBr₆. It still maintains strong absorption. It has favorable carrier effective masses.

Cs₂NiBaI₆ has the largest static reflectivity. Its $R(0)$ is 0.360 (see Table S14). This matches its large $\epsilon_1(0)$ and $n(0)$. However, it has weak near-edge absorption. High polarizability alone does not make it an optimal photovoltaic absorber. Cs₂AgInCl₆ and Cs₂GeSrBr₆ have low static reflectivity. Cs₂AgInCl₆ has $R(0) = 0.131$. Cs₂GeSrBr₆ has $R(0) = 0.100$. Low reflectivity is useful only with strong absorption in the target spectral range.

3.9.5. Energy-loss function and plasmonic optical-loss response

The electron energy-loss function $L(\omega)$ was calculated to evaluate optical dissipation (see Table S15 and Fig. S11). Cs₂SnGeBr₆ shows the most favorable loss behavior for absorber-oriented applications, with comparatively low visible-region loss ($L_{\text{max}} = 0.147$ near 3.18 eV) (see Table S15) combined with a direct band gap, strong absorption, and low effective masses. Rb₂TeCuBr₆ shows moderate visible-region loss. Its L_{max} is 0.371 at 2.60 eV. It also has a strong high-energy loss peak. This suggests that reflection and dissipative losses should be considered during device optimization. Cs₂NiBaI₆ has the strongest high-energy loss peak. This matches its large dielectric constant. However, its indirect band gap and weak near-edge absorption show that high polarizability alone is not enough. Cs₂GeSrBr₆ shows very weak visible-region loss. This matches its wide band gap and limited visible absorption. This may be useful for transparent or UV-selective applications.

The energy-loss analysis supports the optical design hierarchy. This hierarchy comes from the dielectric-function, refractive-index, reflectivity, absorption, and effective-mass results. Cs₂SnGeBr₆ and Rb₂TeCuBr₆ are the strongest absorber-oriented candidates (see Table S15). They have different design trade-offs. Cs₂SnGeBr₆ combines strong absorption with comparatively low visible-region loss, whereas Rb₂TeCuBr₆ provides stronger low-energy light–matter interaction but may require more careful optical-loss management. These results reinforce the main concept of this work: thermodynamically stable lead-free double perovskites should be discovered through backward mapping from device targets to chemical genomes, where absorption strength, dielectric screening, reflectivity loss, plasmonic dissipation, band-edge orbital character, and carrier transport are optimized together rather than treated as isolated descriptors.

3.9.6. Optical-conductivity response and photoinduced charge-excitation behavior

The frequency-dependent optical conductivity, $\sigma(\omega) = \sigma_1(\omega) + i\sigma_2(\omega)$, was analyzed to further evaluate the photoinduced charge-excitation behavior of the DFT-validated lead-free double perovskites. The real component, $\sigma_1(\omega)$, represents the dissipative optical conductivity associated with interband electronic transitions, whereas the imaginary component, $\sigma_2(\omega)$, describes the reactive polarization response of the material. In the context of backward mapping from device targets to chemical genomes, optical conductivity is a useful descriptor because it indicates how effectively a material converts incident photons into electronic excitations within a target spectral window. Therefore, $\sigma(\omega)$ provides a complementary bridge between optical absorption, dielectric screening, and band-edge transport behavior. The extracted optical-conductivity descriptors are summarized in Table S16, and the corresponding spectra are shown in Fig. S12.

As shown in Fig. S12 and summarized in Table S16, the optical-conductivity spectra exhibit strong composition dependence across the DFT-validated chemical space. The most intense $\sigma_1(\omega)$ peaks generally occur at higher photon energies, indicating that the strongest interband charge excitations are located in the UV or high-energy region for several compounds. This trend is consistent with the dielectric-function, absorption, and energy-loss analyses, where high-energy optical transitions dominate in wider-gap or weakly visible-active materials. These

Table 10
DFT-validated chemical-genome design rules for future A_2BX_6 Discovery.

Design Rule	Chemical Meaning	Actionable Inverse-Design Lever	Expected Device Impact
Enforce formability before function	Steric size compatibility defines the boundaries of accessible double-perovskite frameworks.	Constrain the search space to compositions satisfying $0.8 \leq t \leq 1.1$, $\mu \geq 0.41$, and $\tau < 4.18$.	Eliminates structurally implausible candidates early in the screening process and reduces false positives arising from unfeasible forward predictions.
Optimize cohesion within steric boundaries	Balanced metal–halide interactions enhance framework robustness and resistance to decomposition.	Prioritize candidates exhibiting favorable decomposition thermodynamics and positive binary decomposition enthalpies, e.g., $\Delta H > 20 \text{ meV atom}^{-1}$.	Supports thermodynamic accessibility and operational durability comparable to experimentally realized SOTA reference systems.
Tune polarity for anti-bonding band edges	Metal–halide hybridization influences band-edge character and defect energetics. Antibonding band edges are frequently associated with defect-tolerant semiconductors.	Use PDOS analysis to identify compositions exhibiting active metal <i>s/p</i> - and halide <i>p</i> -orbital mixing near the VBM and CBM, indicative of antibonding band-edge states.	May promote defect-tolerant behavior by favoring shallow defect states over deep recombination centers, thereby reducing susceptibility to non-radiative losses.
Deploy polarization for polaron shielding	Macroscopic dielectric screening attenuates Coulombic interactions and carrier scattering in polar lattices.	Select highly polarizable bromide- or iodide-based frameworks to maximize the static dielectric response, $\epsilon_1(0)$.	Is consistent with enhanced Fröhlich-polaron screening, reduced defect-mediated scattering, and potentially prolonged carrier diffusion lengths and carrier lifetimes.
Avoid localized electronic identities for absorber layers	Localized transition-metal <i>d</i> -states can induce carrier localization and transport asymmetry, whereas dispersive post-transition-metal <i>s/p</i> -states facilitate charge transport.	Screen out flat-band electronic topologies and prioritize delocalized post-transition-metal sublattices exhibiting dispersive band edges.	Reduces the likelihood of severe transport bottlenecks and supports low effective masses conducive to balanced ambipolar extraction (e.g., $m_h^* = 0.054 m_0$ in $\text{Cs}_2\text{SnGeBr}_6$).
Evaluate collective losses after light harvesting	Strong absorption alone does not guarantee efficient photon utilization because parasitic optical losses may coexist with high absorption coefficients.	Simultaneously evaluate $\alpha(\omega)$, $R(\omega)$, $L(\omega)$, and $\sigma_1(\omega)$ to quantify optical impedance and dissipative losses.	Provides a more realistic assessment of photon-to-charge conversion potential and guides target-specific device assignment rather than reliance on a single ranking metric.

results emphasize that a favorable electronic band gap alone is not sufficient to define an efficient optoelectronic absorber; the optical transition strength within the target spectral region must also be considered.

Among the investigated compounds, $\text{Rb}_2\text{TeCuBr}_6$ shows the strongest low-energy charge-excitation response, with a visible–NIR σ_1 maximum of 1.801 at 1.53 eV (see Table S16). This feature agrees well with its strong absorption coefficient, high dielectric response, large refractive index, and balanced electron–hole effective masses. From the backward-mapping perspective, $\text{Rb}_2\text{TeCuBr}_6$ therefore satisfies multiple absorber-oriented device targets simultaneously, including efficient low-energy photon-to-charge excitation, strong light–matter interaction, and favorable effective-mass-limited carrier transport. This makes the Rb–Te–Cu–Br chemical motif a particularly interpretable design direction for low-energy optoelectronic absorbers. $\text{Cs}_2\text{SnGeBr}_6$ also exhibits a highly favorable optical-conductivity profile. Its visible-to-near-UV conductivity reaches 2.129 at 3.18 eV, while the global maximum occurs at 13.80 eV (see Table S16). Importantly, this strong conductivity response is accompanied by a direct band gap, high absorption coefficient, relatively low visible-region energy loss, and low carrier effective masses. These combined properties suggest that $\text{Cs}_2\text{SnGeBr}_6$ can support efficient optical excitation and subsequent charge transport, making it another promising absorber-oriented candidate. Chemically, the Sn/Ge–Br framework appears to promote dispersive band-edge states and strong optical transitions, providing a useful design rule for lead-free double-perovskite discovery.

In contrast, $\text{Cs}_2\text{GeSrBr}_6$ and $\text{Cs}_2\text{AgInCl}_6$ display weaker visible-region optical conductivity but pronounced high-energy conductivity peaks. $\text{Cs}_2\text{GeSrBr}_6$ reaches its strongest response at 14.00 eV, while $\text{Cs}_2\text{AgInCl}_6$ shows the largest global conductivity among the series at 14.08 eV (Table S16). This behavior is consistent with their UV-dominant absorption and loss-function responses. Consequently, these compositions are better mapped to UV-active or transparent optoelectronic targets than to broadband photovoltaic absorbers, despite the favorable electron effective mass observed for $\text{Cs}_2\text{AgInCl}_6$. $\text{Cs}_2\text{NiBaI}_6$ illustrates why multi-descriptor validation is essential in target-driven materials discovery. Although it has the smallest electronic band gap in the series, its visible-region optical conductivity is relatively weak, while its strongest conductivity feature appears at 6.27 eV (see Table S16). This indicates that its near-edge optical transitions are not sufficiently strong for efficient visible-light harvesting, likely due to its indirect-gap character and

band-edge orbital selection effects. Therefore, $\text{Cs}_2\text{NiBaI}_6$ should not be prioritized solely on the basis of its narrow gap, even though it exhibits strong dielectric screening and a light hole effective mass.

Overall, the optical-conductivity analysis reinforces the design hierarchy obtained from the absorption coefficient, dielectric function, refractive index, reflectivity, energy-loss spectra, and carrier effective masses. $\text{Rb}_2\text{TeCuBr}_6$ and $\text{Cs}_2\text{SnGeBr}_6$ remain the most promising absorber-oriented candidates because they combine strong optical conductivity in relevant energy windows with favorable transport and screening descriptors. By contrast, $\text{Cs}_2\text{GeSrBr}_6$ and $\text{Cs}_2\text{AgInCl}_6$ are more naturally suited to UV-active or transparent optoelectronic applications. These findings support the central premise of this work: reliable discovery of thermodynamically stable lead-free double perovskites requires backward mapping from device targets to chemical genomes using a coupled descriptor set that includes optical conductivity, absorption strength, dielectric screening, reflection loss, plasmonic dissipation, band-edge orbital character, and carrier transport.

3.10. Quantum mechanical mixing rules and non-trivial crystalline design principles

The ultimate objective of the backward-mapping workflow is not merely to identify a static set of promising lead-free double perovskites, but to decode the coupled compositional principles associated with favorable optoelectronic performance. To this end, the final DFT-validated candidates were mapped back onto the original six-family chemical-genome framework. Within this paradigm, the constituent composition descriptors constitute fundamental chemical “genes,” whereas the calculated structural, electronic, dielectric, optical, and transport properties represent the observable macroscopic phenotypes. This organization reflects the inherently multi-objective nature of solid-state device design. Efficient optoelectronic materials rarely emerge from the optimization of an isolated parameter; instead, viable candidates generally satisfy a coupled hierarchy of structural formability, thermodynamic robustness, chemically plausible charge accommodation, favorable charge transport, and minimized optical dissipation pathways [38].

As illustrated by the multi-phenotype performance map in Fig. 11, no individual compound simultaneously optimizes every descriptor, uncovering highly non-trivial physical tradeoffs that challenge conventional solid-state heuristics. Most notably, the map establishes visual

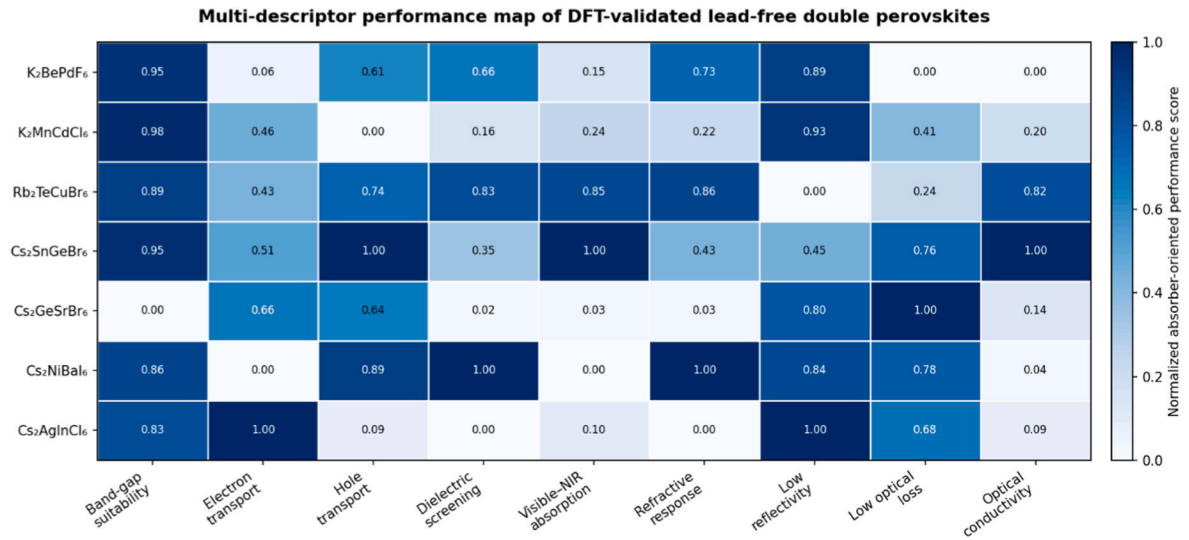


Fig. 11. Multi-phenotype performance map of DFT-validated lead-free double perovskites. Normalized heat map comparing absorber-relevant electronic, transport, dielectric, and optical phenotypes of the DFT-validated candidates. Darker blue indicates a more favorable absorber-oriented score. Rb₂TeCuBr₆ and Cs₂SnGeBr₆ show the most balanced absorber profiles, while Cs₂GeSrBr₆ and Cs₂AgInCl₆ are more naturally suited to UV-active, transparent, or electron-selective optoelectronic targets.

proof of the polarizability-mass asymmetry paradigm in Cs₂SnGeBr₆, where strong electronic screening ($\epsilon_1(0) = 7.54$) coexists with an ultralight hole effective mass ($m_h^* = 0.054 m_0$). These coupled variations are generalized in the schematic decoder in Fig. 12, which illustrates how localized framework cohesion genes (BE_{M-X}) act as a non-linear compensation mechanism capable of stabilizing high-strain, metastable packing manifolds. By shifting the narrative from isolated property tracking to coupled, stability and optoelectronic property trade-offs, these figures define effective-mass-limited transport tendencies and electrostatic screening indicators that remain strictly bounded by ground-state functional constraints while providing an actionable playbook for experimental discovery.

3.10.1. Geometric packing genes define the structural feasibility window

Geometric packing descriptors (t, μ, τ) establish the initial structural boundary for the double-perovskite lattice (Tables 9 and 10). Beyond functioning as a simple pre-filter, steric size matching directly influences downstream electronic properties. Lattice mismatches induce octahedral tilting, local symmetry breaking, and altered $B-X-B'$ bond angles, which modify orbital overlap and alter band dispersion near the band extrema. Constraining candidate selection to established formability windows ($0.8 \leq t \leq 1.1$, $\mu \geq 0.41$, and $\tau < 4.18$) prevents false-positive predictions from structurally implausible configurations.

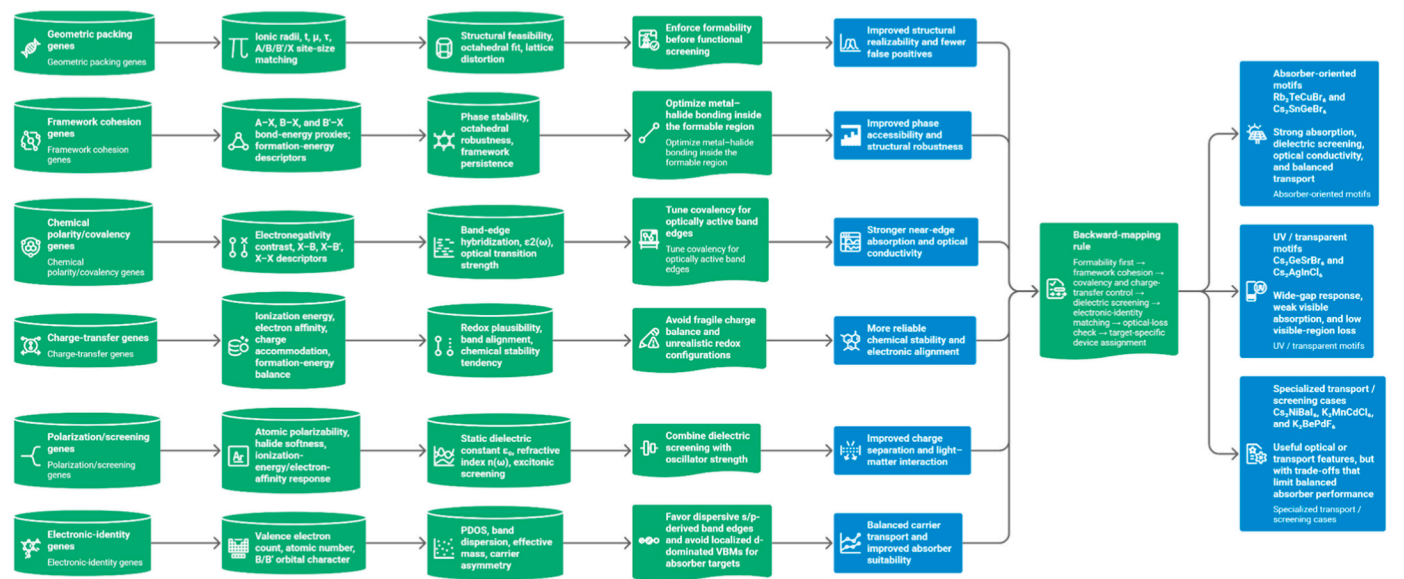


Fig. 12. Schematic overview of the chemical-genome design-rule decoder for lead-free double perovskites. The diagram shows the backward-mapping design logic. It links chemical-gene families to DFT phenotypes and device-level outcomes. Six chemical-gene families are mapped to key device attributes. These families are: geometric packing, framework cohesion, chemical polarity/covalency, charge-transfer, polarization/screening, and electronic-identity genes. The attributes include structural feasibility, phase stability, optical-transition strength, dielectric response, band-edge character, transport behavior, and optical-loss constraints. Absorber-oriented candidates must satisfy multiple criteria. These criteria include formability, cohesion, optically active band edges, dielectric screening, balanced transport, and manageable optical loss.

3.10.2. Framework cohesion genes link stability and electronic robustness

Framework cohesion descriptors, particularly metal-halide bond dissociation energies (BE_{A-X} , BE_{B1-X} , BE_{B3-X}) and atomic formation enthalpies (ΔH_f), track the thermodynamic robustness of the corner-sharing network against competing binary decomposition channels. Strong, balanced metal-halide bonding stabilizes the lattice while influencing the energetic positioning of band-edge states.

This relationship illustrates why single-descriptor screening is insufficient. For instance, while the compact fluoride $K_2\text{BePdF}_6$ exhibits strong dielectric screening, its heavier hole effective mass ($m_h^* = 1.25m_0$) indicates potential transport bottlenecks. In contrast, bromide frameworks like $\text{Rb}_2\text{TeCuBr}_6$ and $\text{Cs}_2\text{SnGeBr}_6$ combine low energy above hull ($\Delta E_{\text{hull}} \leq 0.15$ eV/atom) with dispersive, accessible band structures (Tables 9 and 10).

3.10.3. Chemical polarity and covalency genes tune band-edge hybridization

Chemical polarity and covalency descriptors, represented by electronegativity differences ($\Delta\chi$) and chemical contrast within the octahedral network, influence the degree of orbital mixing at the valence-band maximum (VBM) and conduction-band minimum (CBM). Whereas cohesion descriptors govern lattice stability, polarity-related descriptors modulate quantum-mechanical selection rules and transition probabilities. Our leading absorber candidates, $\text{Cs}_2\text{SnGeBr}_6$ and $\text{Rb}_2\text{TeCuBr}_6$, illustrate this relationship perfectly. Their balanced electronegativity contrasts promote appreciable metal-halide covalency, broadening the electronic band dispersion and reducing carrier effective masses. These hybridization characteristics are also associated with enhanced near-edge optical activity, reflected by high absorption coefficients ($\alpha \sim 10^5 \text{ cm}^{-1}$) and enhanced dynamic optical conductivities ($\sigma_1(\omega)$). Consequently, tuning metal-halide covalency emerges as a useful inverse strategy for enhancing optical response while maintaining structural integrity.

3.10.4. Charge-transfer genes influence chemical plausibility and band-edge positioning

Charge-transfer descriptors incorporate atomic ionization energies (I_A), electron affinities, and the charge-accommodation tendencies of constituent elements. These parameters provide a chemical plausibility filter that helps exclude compositions requiring unstable oxidation states. In addition, such descriptors heavily influence the absolute positioning of the VBM and CBM, as the electronic chemical potentials of the constituent ions contribute directly to the absolute band-edge energies. While explicit heterojunction band alignment calculations fall outside the scope of this work, prioritizing chemically stable charge configurations (B^{2+}/B^{2+} or B^+/B^{3+} states) ensures redox plausibility and increases the likelihood of identifying candidates compatible with practical transport-layer engineering in future device-level investigations.

3.10.5. Polarization and screening genes control dielectric response

Polarization and screening descriptors combine atomic polarizabilities (α), atomic numbers (N), and halide softness to regulate the static dielectric constant $\epsilon_1(0)$ and the frequency-dependent refractive index $n(\omega)$. In polar halide perovskites, macroscopic dielectric screening fundamentally alters carrier-defect interactions and the effectiveness of Coulombic shielding. Heavy halide lattices such as $\text{Cs}_2\text{NiBaI}_6$ and $\text{Rb}_2\text{TeCuBr}_6$ possess highly deformable electron clouds that enhance lattice polarizability. This trend is reflected in the large dielectric constants calculated for $\text{Cs}_2\text{NiBaI}_6$ ($\epsilon_1(0) = 15.28$) and $\text{Rb}_2\text{TeCuBr}_6$ ($\epsilon_1(0) = 13.27$). Within the Fröhlich polaron framework, stronger dielectric screening reduces the Coulombic capture cross-section of deep traps and suppresses excitonic binding tendencies ($\Delta E_{\text{xb}} = -6.43$ meV for $\text{Cs}_2\text{NiBaI}_6$), facilitating free-carrier separation [67]. However, dielectric response alone is not a sufficient predictor of device

performance; as illustrated by K_2BePdF_6 , strong screening cannot completely compensate for heavy transport effective masses, meaning polarization features must be interpreted alongside optical and transport dispersion.

3.10.6. Electronic-identity genes control band-edge character and carrier transport

Electronic-identity descriptors ($V-A$, $V-B$, $N-A$, $N-B$) govern band curvature and transport asymmetry by dictating the specific orbital contributions at the valence band maximum (VBM) and conduction band minimum (CBM). First-principles calculations indicate that delocalized post-transition-metal s - and p -orbitals, such as those in Sn- and Ge-based sublattices, generate dispersive band edges with low carrier effective masses. Conversely, localized transition-metal d -orbitals contract wave-function overlap, producing flat-band topologies and heavier effective masses.

This behavior is illustrated by $\text{Cs}_2\text{NiBaI}_6$, where localized Ni-3d states dominate the VBM, resulting in a large hole effective mass ($m_h^* = 2.45m_0$) despite a light electron branch ($m_e^* = 0.18m_0$). In contrast, $\text{Cs}_2\text{SnGeBr}_6$ exhibits dispersive valence states derived from delocalized Br- p / Sn-Ge hybridization, yielding an ultralight hole effective mass of $m_h^* = 0.054m_0$.

The crucial role of localized orbitals in regulating electronic signatures is a primary focus of several of the designated studies. Sofi [24] and Ahn [69] thoroughly explore the role of 3d transition metal electrons (e.g., V, Mn, Ni) in driving pressure-induced half-metallicity, spin-dependent transport, and 100% spin polarization near the Fermi level in A_2B $B'X_6$ variants. This closely aligns with the fundamental physics discussed by Sofi and Gupta [22] regarding localized f -electron spin configurations in perovskite systems.

While these localized electronic identities are exceptionally valuable for spintronic or thermoelectric applications, such as the vacancy-ordered systems examined by Khan [70] and the cubic bromide networks tracked by Sofi [24], our framework is explicitly trained to treat these localized transition-metal states as electronic transport bottlenecks for solar absorber layers. By contrasting our results with their spin-polarized density of states (DOS) landscapes within Section 3.11.6, we demonstrate how our model uses feature importance to actively screen out these localized d - or f -orbital states in favor of delocalized post-transition-metal s - and p -orbitals (Sn, Ge), thereby avoiding flat-band transport bottlenecks and ensuring low, balanced effective carrier masses.

By incorporating feature importance attributions, the inverse screening funnel actively prioritizes delocalized post-transition-metal s - p orbital interactions over flat d -state configurations. Consequently, compositions with dispersive s - p band edges are identified as favorable candidates for balanced ambipolar charge transport. Computed effective mass descriptors across the validated series are benchmarked against reference systems in Table S20.

3.10.7. Optical-loss phenotypes refine absorber assignment

Optical loss is treated here as a phenotype-level validation layer rather than an independent chemical-gene family. While the absorption coefficient ($\alpha(\omega)$) and the real optical conductivity ($\sigma_1(\omega)$) characterize the efficiency with which photons generate electronic excitations, the reflectivity ($R(\omega)$), extinction coefficient ($k(\omega)$), and energy-loss function ($L(\omega)$) quantify pathways through which incident energy may be dissipated. This multi-dimensional optical assessment reveals important trade-offs among the validated candidates. For example, $\text{Rb}_2\text{TeCuBr}_6$ combines strong visible-to-near-infrared absorption with a high refractive index, leading to increased visible-region reflectivity ($R(0) \sim 0.30$). While this characteristic may increase reflection losses in planar architectures, it also suggests that device-level optimization strategies such as surface texturing or anti-reflection coatings could further recover performance. By comparison, $\text{Cs}_2\text{SnGeBr}_6$ exhibits a more balanced optical

profile, combining strong near-edge absorption with a small visible-region energy-loss maximum ($L_{\text{max}} = 0.147$) and favorable, fluid carrier transport characteristics.

3.10.8. Target-specific assignment of the DFT-validated compounds

Rather than relying on a single composite ranking metric, the backward-mapping framework enables each DFT-validated composition to be associated with the optoelectronic application for which its descriptor profile appears most favorable. For example, $\text{Rb}_2\text{TeCuBr}_6$ exhibits strong visible-NIR absorption, substantial dielectric screening [$\epsilon_1(0) = 13.27$], and relatively balanced carrier effective masses ($m_e^* = 0.32 m_0$, $m_h^* = 0.44 m_0$). These combined characteristics suggest that $\text{Rb}_2\text{TeCuBr}_6$ is a promising candidate for photovoltaic absorbers and broadband photodetectors, particularly when reflection losses can be mitigated through device engineering. Moreover, $\text{Cs}_2\text{SnGeBr}_6$ combines a near-ideal direct band gap (1.112 eV), an ultralow hole effective mass (0.054 m_0), broad visible absorption, and relatively low optical energy loss. Among the materials investigated, it exhibits the most balanced combination of optical and transport characteristics, identifying it as a particularly promising broadband absorber motif. Furthermore, $\text{Cs}_2\text{GeSrBr}_6$, with its wide band gap (3.434 eV), high visible transparency, and minimal low-energy optical losses ($L_{\text{max}} = 0.03$), $\text{Cs}_2\text{GeSrBr}_6$ appears well suited to ultraviolet-selective optoelectronic applications and transparent functional components. Additionally, $\text{Cs}_2\text{AgInCl}_6$ exhibits pronounced transport asymmetry, characterized by an ultralight electron branch ($m_e^* = 0.078 m_0$) and a strongly localized hole branch ($m_h^* = 84.6 m_0$), together with a UV-dominant optical response. These characteristics suggest that its strengths may lie in electron-selective or hole-blocking functionalities rather than in absorber applications requiring balanced ambipolar transport.

3.11. Experimental realization and synthetic roadmaps of targeted endpoints

To bridge the gap between first-principles multi-objective inverse design and laboratory crystallization, we systematically map the realistic synthetic prospects, oxidation-state stability boundaries, and precursor pathways for the prioritized lead-free candidates. A prevalent point of failure in high-throughput database exploration is the selection of compositions that exhibit attractive functional targets but are unfeasible to synthesize due to a high thermodynamic drive toward binary phase separation. To eliminate these false positives, our framework evaluates the total energy of each candidate relative to competing binary decomposition paths (ΔH) derived from foundational structural databases.

As compiled across the structural metrics in Table S19, our primary optoelectronic absorber contender, $\text{Cs}_2\text{SnGeBr}_6$, records a highly favorable convex-hull distance (E_{hull}) of 0.024 eV *atom*⁻¹, matching the thermodynamic access limits of the experimentally synthesized standard inorganic baseline $\text{Cs}_2\text{AgBiBr}_6$ ($E_{\text{hull}} = 0.010$ eV *atom*⁻¹). Concurrently, the remaining prioritized candidates, including $\text{K}_2\text{MnCdCl}_6$ (0.126 eV *atom*⁻¹), $\text{Rb}_2\text{TeCuBr}_6$ (0.129 eV *atom*⁻¹), and $\text{Cs}_2\text{GeSrBr}_6$ (0.150 eV *atom*⁻¹)—all fall safely below or equal to the operational 0.15–0.19 eV *atom*⁻¹ experimental synthesizability ceiling defined by successfully crystallized double-perovskite reference analogues such as $\text{MA}_2\text{AgBiI}_6$ and $\text{MA}_2\text{AgBiBr}_6$.

This structural realizability is further confirmed by evaluating the coordination constraints and oxidation-state stability of the constituent sublattices. The seven discovered endpoints avoid fragile, transient valence states by relying entirely on stable, localized oxidation configurations ($\text{B}^{2+}/\text{B}^{2+}$ or B^+/B^{3+}) that can be targeted through standard mechanochemical, solid-state sintering, or liquid-phase coordination routes: For example, this wide-gap of $\text{Cs}_2\text{GeSrBr}_6$ ultraviolet contender relies on highly stable Ge^{2+} and Sr^{2+} cations. Laboratory crystallization can be targeted via high-temperature vacuum-sealed solid-state reaction

or mechanochemical ball-milling inside an inert moisture-free glovebox (Ar atmosphere) to suppress potential $\text{Ge}^{2+} \rightarrow \text{Ge}^{4+}$ oxidation pathways, combining gastronomic stoichiometric balances of high-purity binary precursors: $2\text{CsBr} + \text{GeBr}_2 + \text{SrBr}_2 \rightarrow \text{Cs}_2\text{GeSrBr}_6$. Moreover, for $\text{Cs}_2\text{NiBaI}_6$ capitalizing on the stable pairing of Ni^{2+} and Ba^{2+} , this highly polarizable narrow-gap framework can be synthesized through standard low-temperature solution crystallization or solvent-evaporation methods utilizing concentrated hydroiodic acid (HI) pathways, dissolving highly soluble CsI , NiI_2 , and BaI_2 baseline precursors under continuous stirring. Again, the compact fluoro-perovskite framework hosts stable Be^{2+} and Pd^{2+} metallic centers for K_2BePdF_6 . Synthesis can realistically be achieved using low-temperature hydrothermal paths inside Teflon-lined autoclaves from concentrated hydrofluoric acid (HF) or high-temperature molten-salt solid-state sintering routines: $2\text{KF} + \text{BeF}_2 + \text{PdF}_2 \rightarrow \text{K}_2\text{BePdF}_6$.

Furthermore, our simulated powder X-ray diffraction (XRD) fingerprints (Table S9) confirm that the fully relaxed crystal structures preserve sharp reflections and weak structural superlattice features characteristic of a high-purity, long-range ordered rock-salt double perovskite lattice. This structural ordering mirrors known experimental structural analogues; for instance, the formability of the germanium-halide sublattices tracks identical geometric distortion spaces as successfully synthesized CsGeI_3 variants, while the chloride framework in $\text{K}_2\text{MnCdCl}_6$ closely aligns with the stable crystallization fields of mixed-metal manganese networks.

These coupled thermodynamic and synthetic boundaries confirm that our genome-guided inverse pipeline restricts its discovery vector to experimentally accessible chemical landscapes.

4. Conclusion

In this study, we implemented a target-driven, backward-mapping discovery framework to identify lead-free $\text{A}_2\text{BB}'\text{X}_6$ double perovskites with balanced thermodynamic and optoelectronic property profiles. By combining charge-balanced chemical space enumeration, geometric formability filtering, a six-family descriptor architecture, evolutionary-optimized surrogate models, and interpretable SHAP analysis, the workflow establishes traceable chemical design rules linking composition to ground-state properties.

Starting from a combinatorial library of 13,088 charge-balanced lead-free formulas, a multi-stage screening funnel systematically narrowed the search space to a select cohort of seven first-principles candidates: K_2BePdF_6 , $\text{K}_2\text{MnCdCl}_6$, $\text{Rb}_2\text{TeCuBr}_6$, $\text{Cs}_2\text{SnGeBr}_6$, $\text{Cs}_2\text{GeSrBr}_6$, $\text{Cs}_2\text{NiBaI}_6$, and $\text{Cs}_2\text{AgInCl}_6$. Density functional theory calculations confirm that these compounds exhibit favorable thermodynamic stability relative to competing binary decomposition channels ($\Delta H > 20$ meV/atom) and maintain thermal lattice persistence during 300 K *ab initio* molecular dynamics (AIMD) simulations.

First-principles evaluation reveals distinct functional profiles across the candidate series rather than a uniform material class. Within single-particle and ground-state dielectric screening models, $\text{Cs}_2\text{SnGeBr}_6$ and $\text{Rb}_2\text{TeCuBr}_6$ emerge as promising absorber-oriented candidates, combining suitable electronic band gaps, light carrier effective masses, and strong static dielectric screening indicators. In contrast, wide-gap frameworks such as $\text{Cs}_2\text{GeSrBr}_6$ and $\text{Cs}_2\text{AgInCl}_6$ map more naturally to ultraviolet-selective or contact-layer functions, while K_2BePdF_6 , $\text{K}_2\text{MnCdCl}_6$, and $\text{Cs}_2\text{NiBaI}_6$ exhibit specialized transport asymmetry.

The central methodological outcome of this work is the chemical-genome rule set: geometric packing descriptors define structural formability; framework cohesion terms govern thermodynamic accessibility; polarity and covalency variables tune band-edge hybridization; charge-transfer descriptors evaluate oxidation state stability; polarization features inform electrostatic screening; and electronic-identity descriptors differentiate dispersive *s*–*p* band edges from flat *d*-state transport bottlenecks.

It is important to emphasize the scope and inherent limits of these computational findings. The property assignments presented here rely on ground-state GGA-PBE calculations, single-particle band structures, and 0 K/300 K computational stability metrics. To bridge these theoretical screening proxies with practical application, future experimental work is necessary to achieve phase-pure laboratory synthesis, characterize native defect formation, evaluate ambient chemical stability against moisture and oxygen, and measure actual carrier mobilities and device power conversion efficiencies. Nevertheless, this target-to-chemistry framework provides an interpretable and reusable strategy for guiding the rational discovery of sustainable lead-free double perovskites.

CRediT authorship contribution statement

Nafis Ahtasum: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Sohanur Rahman Sohan:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Validation, Visualization, Writing – original draft. **Md Mostaq Ahmed Himel:** Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Visualization, Writing – original draft. **Md Zahid Hassan:** Formal analysis, Investigation, Software, Visualization, Writing – original draft. **Muhammad Harussani Moklis:** Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **Md Rafiul Alam Roni:** Project administration, Resources, Software, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mtphys.2026.102180>.

Data availability

Data will be made available on request.

References

- Y. Wei, J. He, C. Yang, W. Yu, J. Feng, X.-J. Liu, X. Chong, Accelerated Multi-Property Screening of Lead-Free Halide Double Perovskite via Transfer Learning, 2025, <https://doi.org/10.1002/adfm.202514377>.
- J. Schmidt, J. Shi, P. Borlido, L. Chen, S. Botti, M.A.L. Marques, Predicting the thermodynamic stability of solids combining density functional theory and machine learning, *Chem. Mater.* 29 (2017) 5090–5103, <https://doi.org/10.1021/acs.chemmater.7b00156>.
- T. Lu, D. Cortie, Z.-X. Li, N. Narayanan, Z. Liu, Q. Sun, T.J. Frankcombe, G. J. McIntyre, D. Yu, Y. Liu, Role of A-Site molecular ions in the polar functionality of metal–organic framework perovskites, *Chem. Mater.* 33 (2021) 9666–9676, <https://doi.org/10.1021/acs.chemmater.1c03276>.
- G. Pilania, A. Mannodi-Kanakkithodi, B.P. Uberuaga, R. Ramprasad, J. E. Gubernatis, T. Lookman, Machine learning bandgaps of double perovskites, *Sci. Rep.* 6 (2016): 19375, <https://doi.org/10.1038/srep19375>.
- C. Chen, W. Ye, Y. Zuo, C. Zheng, S.P. Ong, Graph networks as a universal machine learning framework for molecules and crystals, *Chem. Mater.* 31 (2019) 3564–3572, <https://doi.org/10.1021/acs.chemmater.9b01294>.
- E. Landini, K. Reuter, H. Oberhofer, Machine-Learning Based Screening of Lead-free Halide Double Perovskites for Photovoltaic Applications, 2022, <https://doi.org/10.48550/arXiv.2208.12736>.
- Z. Chen, J. Wang, C. Li, B. Liu, D. Luo, Y. Min, N. Fu, Q. Xue, Highly versatile and accurate machine learning methods for predicting perovskite properties, *J. Mater. Chem. C* 12 (2024) 15444–15453, <https://doi.org/10.1039/D4TC02268H>.
- Z. Gao, G. Mao, S. Chen, Y. Bai, P. Gao, C. Wu, I.D. Gates, W. Yang, X. Ding, J. Yao, High throughput screening of promising lead-free inorganic halide double perovskites via first-principles calculations, *Phys. Chem. Chem. Phys.* 24 (2022) 3460–3469, <https://doi.org/10.1039/D1CP04976C>.
- M.Y. Sofi, M.S. Khan, J. Ali, M.A. Khan, Exploring the lead-free halide Cs₂MgBaBr₆ (M = Li, Na) double perovskites for sustainable energy applications, *Sci. Rep.* 14 (2024) 5520, <https://doi.org/10.1038/s41598-024-54386-1>.
- B. Liu, H. Zhou, Z. Chen, J. Li, X. Fang, Z. Chen, X. Liu, H. Liu, L. Xiao, J. Lin, Spectroscopic limited maximum efficiency-first progressive screening of halide double perovskites via first principle-constrained machine learning, *ACS Appl. Energy Mater.* 9 (2026) 6757–6771, <https://doi.org/10.1021/acsaem.6c00044>.
- D. Bartsaghi, A.H. Slavney, M.C. Gélvez-Rueda, B.A. Connor, F.C. Grozema, H. I. Karunadasa, T.J. Savenije, Charge carrier dynamics in Cs₂AgBiBr₆ double perovskite, *J. Phys. Chem. C* 122 (2018) 4809–4816, <https://doi.org/10.1021/acs.jpcc.8b00572>.
- M.Y. Sofi, M.S. Khan, M.A. Khan, Eco-friendly lead-free halide double perovskites A₂CuMCl₆ (A = K, Rb; M = Sb, Bi): stability, thermoelectric, and optoelectronic advancements through theoretical insights, *J. Mater. Chem. C* 12 (2024) 16045–16058, <https://doi.org/10.1039/D4TC00891J>.
- Z. He, S. Zhou, H. Zhu, J. Han, Y. Hu, C. Shen, Y. Cheng, D. Yang, M. Zhang, Z. Shao, Q. Zhang, X. Mo, Advancing wide-band-gap perovskite solar cells via vapor-phase deposition, *Cell Rep. Phys. Sci.* 7 (2026), <https://doi.org/10.1016/j.xcrp.2026.103346>.
- W. Paritmongkol, N.S. Dahod, A. Stollmann, N. Mao, C. Settens, S.-L. Zheng, W. A. Tisdale, Synthetic variation and structural trends in layered two-dimensional alkylammonium lead halide perovskites, *Chem. Mater.* 31 (2019) 5592–5607, <https://doi.org/10.1021/acs.chemmater.9b01318>.
- W. Travis, E.N.K. Glover, H. Bronstein, D.O. Scanlon, R.G. Palgrave, On the application of the tolerance factor to inorganic and hybrid halide perovskites: a revised system, *Chem. Sci.* 7 (2016) 4548–4556, <https://doi.org/10.1039/C5SC04845A>.
- S. Djaili, M. Hachi, B. Lagoun, T. Smain, A. Amor, K. Souleh, First-principles study of electronic and optical properties of Cd_xMg_{1-x} – xO alloys, TB-mBJ calculations, *Opt. Quant. Electron.* 55 (2023) 297, <https://doi.org/10.1007/s11082-023-04612-3>.
- M. Baskurt, P. Erhart, J. Wiktor, Direct, indirect, and self-trapped excitons in Cs₂AgBiBr₆, *J. Phys. Chem. Lett.* 15 (2024) 8549–8554, <https://doi.org/10.1021/acs.jpclett.4c01604>.
- S.N. Rahane, G.K. Rahane, A. Mandal, Y. Jadhav, A. Godha, A. Rokade, S. Shah, Y. Hase, A. Waghmare, N.G. Saykar, A. Roy, K.N. Salgaonkar, D. Dubal, S. K. Makineni, N.Y. Dzade, S.R. Jadhav, S.R. Rondiya, Lead-Free Cs₂AgBiCl₆ double perovskite: experimental and theoretical insights into the self-trapping for optoelectronic applications, *ACS Phys. Chem. Au* 4 (2024) 476–489, <https://doi.org/10.1021/acsphyschemau.4c00008>.
- A. Qazi, A.A.K. Jadoon, Explainable ensemble learning using SHAP for ERP anomaly detection, *Sci. Rep.* (2026), <https://doi.org/10.1038/s41598-026-57913-4>.
- F. Musil, A. Grisafi, A.P. Bartók, C. Ortner, G. Csányi, M. Ceriotti, Physics-Inspired structural representations for molecules and materials, *Chem. Rev.* 121 (2021) 9759–9815, <https://doi.org/10.1021/acs.chemrev.1c00021>.
- G.O. Kayode, M.M. Montemore, Latent variable machine learning framework for catalysis: general models, transfer learning, and interpretability, *JACS Au* 4 (2024) 80–91, <https://doi.org/10.1021/jacsau.3c00419>.
- M.Y. Sofi, D.C. Gupta, Scrutinized the inherent spin half-metallicity and thermoelectric response of f-electron-based RbMO₃ (M = Np, Pu) perovskites: a computational assessment, *Sci. Rep.* 12 (2022): 19476, <https://doi.org/10.1038/s41598-022-22633-y>.
- B.P. Bloom, Y. Paltiel, R. Naaman, D.H. Waldeck, Chiral induced spin selectivity, *Chem. Rev.* 124 (2024) 1950–1991, <https://doi.org/10.1021/acs.chemrev.3c00661>.
- M.Y. Sofi, M.S. Khan, M.A. Khan, Semiconducting ferromagnetism and thermoelectric performance of Rb₂GeM₆ (M = V, Ni, Mn): a computational perspective, *Mater. Adv.* 6 (2025) 2071–2089, <https://doi.org/10.1039/D4MA01091D>.
- M.Y. Sofi, M.S. Khan, J. Ali, M.A. Khan, Unlocking the role of 3d electrons on ferromagnetism and spin-dependent transport properties in K₂GeNiX₆ (X=Br, I) for spintronics and thermoelectric applications, *J. Phys. Chem. Solid.* 192 (2024): 112022, <https://doi.org/10.1016/j.jpcs.2024.112022>.
- M.Y. Sofi, M.S. Khan, J. Ali, M.A. Khan, Thermoelectric response of vacancy order double perovskite K₂PtBr₆ for energy harvesting applications, in: R.K. Kotnala, A. Sharma Kaushik, S.S. Subramanian, A.K. Vishwakarma (Eds.), *Adv. Funct. Mater. Sustain. Environ.*, Springer Nature Switzerland, Cham, 2024, pp. 179–188, https://doi.org/10.1007/978-3-031-62620-3_17.
- M. Kulichenko, B. Nebgen, N. Lubbers, J.S. Smith, K. Barros, A.E.A. Allen, A. Habib, E. Shinkle, N. Fedik, Y.W. Li, R.A. Messerly, S. Tretiak, Data generation for machine learning interatomic potentials and beyond, *Chem. Rev.* 124 (2024) 13681–13714, <https://doi.org/10.1021/acs.chemrev.4c00572>.
- M. Otyepka, M. Pykal, M. Otyepka, Advancing materials discovery through artificial intelligence, *Appl. Mater. Today* 47 (2025): 102981, <https://doi.org/10.1016/j.apmt.2025.102981>.
- Y.-J. Choi, J.-J. Lee, J.-S. Park, H. Kang, M. Kim, J. Kim, D. Okada, D.H. Kim, F. Araoka, S.-W. Choi, Circularly polarized light emission from Nonchiral perovskites incorporated into nanoporous cholesteric polymer templates, *ACS Nano* 18 (2024) 909–918, <https://doi.org/10.1021/acsnano.3c09596>.

- [30] Z. Pan, L. Wei, T. Wang, Z. Zhao, H. Li, R. Zhao, Machine learning for functional materials: insights from compositional, electronic, and dynamic processes, *AAPPS Bull.* 36 (2026) 5, <https://doi.org/10.1007/s43673-026-00185-x>.
- [31] C.J. Taylor, A. Pomberger, K.C. Felton, R. Grainger, M. Barecka, T.W. Chamberlain, R.A. Bourne, C.N. Johnson, A.A. Lapkin, A brief introduction to chemical reaction optimization, *Chem. Rev.* 123 (2023) 3089–3126, <https://doi.org/10.1021/acs.chemrev.2c00798>.
- [32] A. Talapatra, B.P. Ueberuaga, C.R. Stanek, G. Pilania, Band gap predictions of double perovskite oxides using machine learning, *Commun. Mater.* 4 (2023) 46, <https://doi.org/10.1038/s43246-023-00373-4>.
- [33] T. Gong, J. Zhou, X. Wang, Interpretable machine learning screening and analysis of pressure sensitivity and high-pressure emission line shifts in lanthanide- and transition-metal-doped phosphors, *J. Mater. Chem. C* (2026), <https://doi.org/10.1039/D6TC00745G>.
- [34] C.P. Egwuogu, N.V. Ogueke, E.E. Oguzie, O.K. Echendu, A.G. Boldyreva, M. M. Tepliakova, C.A. Madu, Electronic and optical properties of mixed-halide perovskites, *Cs_xFA_{1-x}Pb(X, X')₃*: insights from density functional theory, *Discov. Electron.* 3 (2026) 31, <https://doi.org/10.1007/s44291-026-00182-2>.
- [35] F. Fahim, M. Ramzan, M.W. Mukhtar, Z. Nazeer, M.N. Akhtar, K.M. Katubi, M. S. Al-Buriah, S. Ullah, DFT insights of the physical, optoelectronic, and energy loss function features of Cd-doped Bi₂O₃ oxide materials for photocatalytic applications, *J. Mol. Graph. Model.* 137 (2025): 109019, <https://doi.org/10.1016/j.jmgm.2025.109019>.
- [36] Y. Jin, X. Wang, M. Yao, D. Qiu, D.J. Singh, J. Xi, J. Yang, L. Xi, High-throughput deformation potential and electrical transport calculations, *npj Comput. Mater.* 9 (2023) 190, <https://doi.org/10.1038/s41524-023-01153-x>.
- [37] D.C. Hiener, G.R. Hutchison, Pareto optimization of oligomer polarizability and dipole moment using a genetic algorithm, *J. Phys. Chem. A* 126 (2022) 2750–2760, <https://doi.org/10.1021/acs.jpca.2c01266>.
- [38] E. Landini, K. Reuter, H. Oberhofer, Machine-Learning Based Screening of Lead-free Halide Double Perovskites for Photovoltaic Applications, 2022, <https://doi.org/10.48550/arXiv.2208.12736>.
- [39] A. Merchant, S. Batzner, S.S. Schoenholz, M. Aykol, G. Cheon, E.D. Cubuk, Scaling deep learning for materials discovery, *Nature* 624 (2023) 80–85, <https://doi.org/10.1038/s41586-023-06735-9>.
- [40] S. Gao, Q. Huang, C. Huang, C. Li, K. Liu, B. Sa, Y. Yu, D. Xue, Z. Liu, M. Dai, Deep generative model for the inverse design of Van der Waals heterostructures, *Sci. Rep.* 15 (2025): 23023, <https://doi.org/10.1038/s41598-025-06432-9>.
- [41] L. Ward, A. Agrawal, A. Choudhary, C. Wolverton, A general-purpose machine learning framework for predicting properties of inorganic materials, *npj Comput. Mater.* 2 (2016): 16028, <https://doi.org/10.1038/npjcompumats.2016.28>.
- [42] S. Curtarolo, G.L.W. Hart, M.B. Nardelli, N. Mingo, S. Sanvito, O. Levy, The high-throughput highway to computational materials design, *Nat. Mater.* 12 (2013) 191–201, <https://doi.org/10.1038/nmat3568>.
- [43] A. Jain, S.P. Ong, G. Hautier, W. Chen, W.D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, K.A. Persson, Commentary: the Materials project: a materials genome approach to accelerating materials innovation, *APL Mater.* 1 (2013): 011002, <https://doi.org/10.1063/1.4812323>.
- [44] J. Lin, J. Fukuyama, A Comparative Analysis of Machine Learning Models in SHAP Analysis, 2026, <https://doi.org/10.48550/arXiv.2604.07258>.
- [45] S.J. Wetzel, S. Ha, R. Iten, M. Klopotek, Z. Liu, Interpretable Machine Learning in Physics: a Review, 2025, <https://doi.org/10.48550/arXiv.2503.23616>.
- [46] A.T. Keleko, B. Kamsu-Foguem, R.H. Ngouna, A. Tongne, Health condition monitoring of a complex hydraulic system using Deep Neural Network and DeepSHAP explainable XAI, *Adv. Eng. Software* 175 (2023): 103339, <https://doi.org/10.1016/j.advengsoft.2022.103339>.
- [47] J.S. Manser, J.A. Christians, P.V. Kamat, Intriguing optoelectronic properties of metal halide perovskites, *Chem. Rev.* 116 (2016) 12956–13008, <https://doi.org/10.1021/acs.chemrev.6b00136>.
- [48] V.A. Osipov, B. Fainberg, Exciton pairs coupled via long-living phonons and their superfluorescent markers, *Phys. Rev. B* 112 (2025): 235206, <https://doi.org/10.1103/PhysRevB.112.235206>.
- [49] Z. Muhammad, A. Rashid, Exciton binding energies and polaron interplay in the optically excited state of organic–inorganic lead halide perovskites, *Mater. Adv.* 6 (2025) 13–38, <https://doi.org/10.1039/D4MA00454J>.
- [50] Z. Guo, B. Lin, Machine learning stability and band gap of lead-free halide double perovskite materials for perovskite solar cells, *Sol. Energy* 228 (2021) 689–699, <https://doi.org/10.1016/j.solener.2021.09.030>.
- [51] X.-G. Zhao, J.-H. Yang, Y. Fu, D. Yang, Q. Xu, L. Yu, S.-H. Wei, L. Zhang, Design of lead-free inorganic halide perovskites for solar cells via cation-transmutation, *J. Am. Chem. Soc.* 139 (2017) 2630–2638, <https://doi.org/10.1021/jacs.6b09645>.
- [52] A. Ullah, M. Iftikhar Khan, Ihtisham-ul-haq, B.S. Almutairi, D.B.N. AlResheedi, J. R. Choi, Bandgap engineering and enhancing optoelectronic performance of a lead-free double perovskite Cs₂AgBiBr₆ solar cell via Al Doping, *ACS Omega* 9 (2024) 18202–18211, <https://doi.org/10.1021/acsomega.3c10388>.
- [53] Z. Zhang, Q. Sun, Y. Lu, F. Lu, X. Mu, S.-H. Wei, M. Sui, Hydrogenated Cs₂AgBiBr₆ for significantly improved efficiency of lead-free inorganic double perovskite solar cell, *Nat. Commun.* 13 (2022) 3397, <https://doi.org/10.1038/s41467-022-31016-w>.
- [54] L.M. Wheeler, V.M. Wheeler, Detailed balance analysis of photovoltaic windows, *ACS Energy Lett.* 4 (2019) 2130–2136, <https://doi.org/10.1021/acsenergylett.9b01316>.
- [55] J. Chen, X. Ma, L. Gong, C. Zhou, J. Chen, Y. Lu, M. Zhou, H. He, Z. Ye, Improving the performance of lead-free Cs₂AgBiBr₆ double perovskite solar cells by passivating Br vacancies, *J. Mater. Chem. C* 12 (2024) 14074–14084, <https://doi.org/10.1039/D4TC02339K>.
- [56] R.-I. Biega, H.J. Jöbsis, Z. Gijssels, M. Hüskens, E.M. Hutter, L. Leppert, Halide mixing in Cs₂AgBi(I x Br 1–x) double perovskites: a pathway to tunable excitonic properties, *J. Phys. Chem. C* 128 (2024) 14767–14775, <https://doi.org/10.1021/acs.jpcc.4c04453>.
- [57] Z. Ran, X. Wang, Y. Li, D. Yang, X.-G. Zhao, K. Biswas, D.J. Singh, L. Zhang, Bismuth and antimony-based oxyhalides and chalcogenides as potential optoelectronic materials, *npj Comput. Mater.* 4 (2018) 14, <https://doi.org/10.1038/s41524-018-0071-1>.
- [58] S.N. Rahane, G.K. Rahane, A. Mandal, Y. Jadhav, A. Godha, A. Rokade, S. Shah, Y. Hase, A. Waghmare, N.G. Saykar, A. Roy, K.N. Salgaonkar, D. Dubal, S. K. Makineni, N.Y. Dzade, S.R. Jadhav, S.R. Rondiya, Lead-Free Cs₂AgBiCl₆ double perovskite: experimental and theoretical insights into the self-trapping for optoelectronic applications, *ACS Phys. Chem. Au* 4 (2024) 476–489, <https://doi.org/10.1021/acspchemau.4c00008>.
- [59] M.B. Bechir, S. Znaidia, Understanding the polaron behavior in Cs₂CuSbCl₆ halide double perovskite, *Phys. Chem. Chem. Phys.* 25 (2023) 19684–19692, <https://doi.org/10.1039/D3CP02176A>.
- [60] Y. Cui, X.-Y. Liu, X.-F. Ma, J.-P. Deng, Y.-Y. Liu, Z.-Q. Li, Z.-W. Wang, Infrared Optical Absorption of Fr Ohlrich Polaron in Metal Halide Perovskites, 2022, <https://doi.org/10.48550/arXiv.2201.06360>.
- [61] I.O. Simonenko, R.G. Nazmitdinov, V.A. Kinev, A.R. Tameev, A. Moia-Pol, Halide perovskite scintillators for X-ray detection: from structure to engineering, *Phys. Chem. Chem. Phys.* 28 (2026) 4992–5042, <https://doi.org/10.1039/D5CP03941J>.
- [62] J. Zhang, B. Che, W. Zhao, Y. Fang, R. Han, Y. Yang, J. Liu, T. Yang, T. Chen, N. Yuan, J. Ding, S. (Frank) Liu, Polar species for effective dielectric regulation to achieve high-performance CsPbI₃ solar cells, *Adv. Mater.* 34 (2022): 2202735, <https://doi.org/10.1002/adma.202202735>.
- [63] M. Delor, A.H. Slavney, N.R. Wolf, M.R. Filip, J.B. Neaton, H.I. Karunadasa, N. S. Ginsberg, Carrier diffusion lengths exceeding 1 μm despite trap-limited transport in halide double perovskites, *ACS Energy Lett.* 5 (2020) 1337–1345, <https://doi.org/10.1021/acsenergylett.0c00414>.
- [64] N. Li, Q. Ma, Y. Zhao, Y. Qiu, First-Principles Study on stability and photoelectrical properties of hybrid perovskite systems FA₂BBr_{4-x}l_x (B = Pb, Sn, ge; x = 0, 1, 2, 3, 4), *J. Phys. Chem. C* 128 (2024) 10290–10299, <https://doi.org/10.1021/acs.jpcc.4c01262>.
- [65] J.E. Ingall, E. Linscott, N. Colonna, A.J. Page, V.J. Keast, Accurate and efficient computation of the fundamental bandgap of the vacancy-ordered double Perovskite Cs₂TiBr₆, *J. Phys. Chem. C* 128 (2024) 9217–9228, <https://doi.org/10.1021/acs.jpcc.3c07957>.
- [66] S.A. Tawfik, M. Rashid, S. Gupta, S.P. Russo, T.R. Walsh, S. Venkatesh, Machine learning-based discovery of vibrationally stable materials, *npj Comput. Mater.* 9 (2023) 5, <https://doi.org/10.1038/s41524-022-00943-z>.
- [67] M.S. Shadabroo, H. Abdizadeh, M.R. Golobostanfard, Elpasolite structures based on A₂AgBiX₆ (A: MA, Cs, X: i, Br): application in double perovskite solar cells, *Mater. Sci. Semicond. Process.* 125 (2021): 105639, <https://doi.org/10.1016/j.mssp.2020.105639>.
- [68] J. Du, C. Yang, B. Zhu, C. Tian, J. Shi, C. Xia, Anion-engineered chalcogenide perovskites Ba₂HfCh₄: from light emitters to photovoltaics, *J. Mater. Chem. A* 14 (2026) 14556–14566, <https://doi.org/10.1039/D5TA09959E>.
- [69] A. Annaberdiyev, S. Mandal, L. Mitás, J.T. Krogel, P. Ganesh, The role of electron correlations in the electronic structure of putative Chern magnet TbMn₆Sn₆, *npj Quantum Mater.* 8 (2023) 50, <https://doi.org/10.1038/s41535-023-00583-6>.
- [70] M.Y. Sofi, MohdS. Khan, M.A. Khan, Harnessing the half-metallicity and thermoelectric insights in Cs₂AgMBr₆ (M = V, Mn, Ni) double halide perovskites: a DFT study, *Mater. Sci. Semicond. Process.* 186 (2025): 109023, <https://doi.org/10.1016/j.mssp.2024.109023>.